- Jose Luis Saul Vicario,

Estructura:

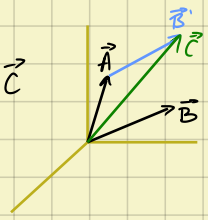
1. Análisis Vectorial y Funcional	25%	80% teoría
2. Ecuaciones Diferenciales Parciales y Resolución	25%	
3. Funciones Especiales Angulares	25%	20% taller
4. Polinomios Ortogonales	25%	

Textos:
- Arfken y Weber
- Alonso Sepúlveda
- P.M. Morse y F. Feshback

1. Coordenadas Curvilíneas

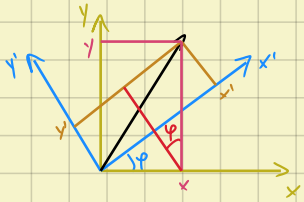
Vectores

$\vec{A} + \vec{B} = \vec{C}$



★ repasar diagonalización

- base $\{\hat{e}_i\} \in \mathbb{R}^n$, escribimos $\vec{v} = \sum_{i=1}^{n} v_i \hat{e}_i$, $v_i = \hat{e}_i \cdot \vec{v}$
- base $\{\hat{e}'_j\} \in \mathbb{R}^n$, escribimos $\vec{v} = \sum_{j=1}^{n} v'_j \hat{e}'_j$, $v'_j = \hat{e}'_j \cdot \vec{v}$



Ahora, $v_i = \hat{e}_i \cdot \vec{v} = \hat{e}_i \cdot \sum_j v'_j \hat{e}'_j = \sum_j v'_j \underbrace{\hat{e}_i \cdot \hat{e}'_j}_{M_{ij}}$

$\therefore v_i = \sum_j M_{ij} v'_j$

↳ $\mathbb{V} = \mathbb{M}\mathbb{V}' \rightarrow \mathbb{V}' = \mathbb{M}^{-1}\mathbb{V}$

Una función puede tomar matrices como argumentos: $e^{Ad} = \begin{pmatrix} e^{a_1} & & 0 \\ & e^{a_2} & \\ 0 & & e^{a_3} \end{pmatrix}$

Para una matriz densa,

$Ad = u^{\dagger} A u \rightarrow u Ad u^{\dagger} = A$ → diagonalizamos

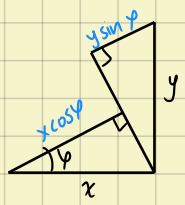
↳ $e^{A} = e^{uAdu^{\dagger}}$ → expandimos en serie de Taylor

pero $e^{A} = 1 + A + \frac{A^2}{2} + \cdots \rightarrow u^{\dagger} e^{A} u = 1 + Ad + \frac{Ad^2}{2} + \cdots = e^{Ad}$

$\rightarrow e^{A} = u e^{Ad} u^{\dagger}$

En general, esto se cumple para toda función analítica

Del dibujo,

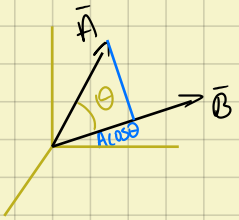


$x' = x\cos\varphi + y\sin\varphi$
$y' = -x\sin\varphi + y\cos\varphi$
$\rightarrow \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\varphi & \sin\varphi \\ -\sin\varphi & \cos\varphi \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

★ para multiplicar matrices
$\mathbb{C} = A \cdot B = \sum_j a_{ij} b_{jk} = C_{ik}$

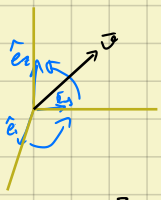
$\mathbb{D} = A\,B\,C = \sum_j \sum_l a_{ij} b_{jl} c_{lk} = d_{ik}$

$$A_x = \hat{e}_x \cdot \vec{A} = A\cos\gamma$$
$$A_y = \hat{e}_y \cdot \vec{A} = A\cos\beta \quad \rightarrow \quad \hat{e}_i \cdot \hat{e}_j = \delta_{ij} \text{ (ortonormal)}$$
$$A_z = \hat{e}_z \cdot \vec{A} = A\cos\alpha$$



$$\vec{A} \cdot \vec{B} = A_B \, B = AB\cos\theta$$

Notación Tensorial



$$\hat{e}_i \times \hat{e}_j = \varepsilon_{ijk}\hat{e}_k \rightarrow \text{sistema dextrógiro}$$
$$\vec{v} = v_i\hat{e}_i = \sum_{i=1}^{n} v_i\hat{e}_i \in \mathbb{R}^3$$

$$\rightarrow \vec{x}\cdot\vec{y} = (x_i\hat{e}_i)(y_j\hat{e}_j) = x_i y_j \, \hat{e}_i\cdot\hat{e}_j = x_i y_j \delta_{ij} = x_i y_i$$

$$\vec{x}\times\vec{y} = x_i\hat{e}_i \times y_j\hat{e}_j = x_i y_j (\hat{e}_i \times \hat{e}_j) = x_i y_j \varepsilon_{ijk}\hat{e}_k \rightarrow \text{los índices son mudos} = \hat{e}_j \varepsilon_{jik} x_i y_k = \hat{e}_i \varepsilon_{ijk} x_j y_k$$

- Tensor de Levi-Civita

$$\varepsilon_{ijk} = \begin{cases} +1 & \text{si } i,j,k \text{ (cíclico)} \\ -1 & \text{si } i,k,j \text{ (cíclico)} \\ 0 & \text{si se repiten índices} \end{cases}$$

$$\vec{L} = \vec{r}\times\vec{p} = \varepsilon_{ijk}\, x_i p_j \hat{e}_k$$

Mostremos $\vec{A}\times(\vec{B}\times\vec{C}) = \vec{B}(\vec{A}\cdot\vec{C}) - \vec{C}(\vec{A}\cdot\vec{B})$

$$\vec{A} = a_j\hat{e}_j$$
$$\vec{B}\times\vec{C} = b_k c_l \, \varepsilon_{klm}\hat{e}_m$$
$$[\vec{A}\times(\vec{B}\times\vec{C})]_i = [a_j\hat{e}_j \times b_k c_l \varepsilon_{klm}\hat{e}_m]_i = [a_j b_k c_l \, \varepsilon_{klm}\underbrace{(\hat{e}_j\times\hat{e}_m)}_{\varepsilon_{mji}\hat{e}_i}]_i$$
$$= a_j b_k c_l \, \underbrace{\varepsilon_{klm}\varepsilon_{mji}}_{(\delta_{ki}\delta_{lj} - \delta_{kj}\delta_{li})}\hat{e}_i = (a_l b_i c_l - a_k b_k c_i)\hat{e}_i = b_i(a_l c_l)\hat{e}_i - c_i(a_k b_k)\hat{e}_i$$
$$= \vec{B}(\vec{A}\cdot\vec{C}) - \vec{C}(\vec{A}\cdot\vec{B})$$

Algunas propiedades

- $\vec{A}\times(\vec{B}\times\vec{C}) = \vec{B}(\vec{A}\cdot\vec{C}) - \vec{C}(\vec{A}\cdot\vec{B})$
- $\vec{\nabla}\times\vec{\nabla}\times\vec{A} = \nabla(\vec{\nabla}\cdot\vec{A}) - (\vec{\nabla}\cdot\vec{\nabla})\vec{A}$
- $\vec{A}\times(\vec{\nabla}\times\vec{A}) = \frac{1}{2}\nabla(\vec{A}^2) - (\vec{A}\cdot\vec{\nabla})\vec{A}$
- $\vec{A}\times(\vec{B}\times\vec{C}) + \vec{B}\times(\vec{C}\times\vec{A}) + \vec{C}\times(\vec{A}\times\vec{B}) = 0$
- rot (grad) $\rightarrow \vec{\nabla}\times\underbrace{\vec{\nabla}\varphi}_{\text{vector irrotacional}} = 0$
- div (rot) $\rightarrow \vec{\nabla}\cdot(\underbrace{\vec{\nabla}\times\vec{\nabla}}_{\text{vector solenoidal}}) = 0$

Gradientes

$$f(x+\Delta x) = f(x) + \left.\frac{df}{dx}\right|_x \Delta x$$
$$f(\vec{r}+\Delta\vec{r}) = f(\vec{r}) + \nabla f|_{\vec{r}}\,\Delta\vec{r} + \dots$$
$$= f(\vec{r}) + df \rightarrow df = \nabla f|_{\vec{r}}\cdot d\vec{r}$$
$$\nabla(u\cdot v) = v\nabla u + u\nabla v$$

PREGUNTA PARCIAL

Divergencia

$$\vec{\nabla}\cdot\vec{V} = \left(\frac{\partial}{\partial x_i}\right)(V_i) = \frac{\partial V_{x_1}}{\partial x_1} + \frac{\partial V_{x_2}}{\partial x_2} + \cdots + \frac{\partial V_{x_n}}{\partial x_n}$$

En las eq. de Maxwell,

- $\vec{\nabla}\cdot\vec{B} = 0 \rightarrow \vec{B} = \vec{\nabla}\times\vec{A}$ y $\vec{\nabla}(\vec{\nabla}\times\vec{A}) = 0$
- $\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} = -\frac{\partial}{\partial t}(\vec{\nabla}\times\vec{A})$

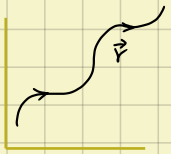
$\rightarrow \vec{\nabla}\times\vec{E} + \frac{\partial}{\partial t}(\vec{\nabla}\times\vec{A}) = \vec{\nabla}\times\vec{E} + \vec{\nabla}\times\frac{\partial\vec{A}}{\partial t} = \vec{\nabla}\times\left(\vec{E} + \frac{\partial\vec{A}}{\partial t}\right) = 0$ si y solo si $\vec{E} + \frac{\partial\vec{A}}{\partial t} = -\vec{\nabla}\phi$

$\rightarrow \vec{E} = -\vec{\nabla}\phi - \frac{\partial\vec{A}}{\partial t}$

Integrales

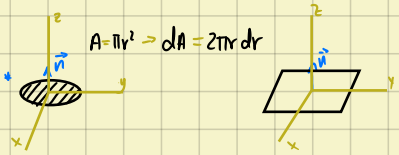
1. Línea:

- $\int \varphi(x,y,z)\, d\vec{r} = \hat{x}\int \varphi\, dx + \hat{y}\int \varphi\, dy + \hat{z}\int \varphi\, dz$
- $\int \vec{F}(x,y,z)\, d\vec{r} = \int (F_x, F_y, F_z)\cdot(dx, dy, dz) = \int F_x dx + \int F_y dy + \int F_z dz$



2. Superficie:

$$d\vec{S} = \vec{n}\, dA$$



* se define de forma dextrógira

- $\int \varphi(x,y,z)\cdot d\vec{S} = \vec{n}\cdot\int \varphi(x,y,z)\, dA$
- $\int \vec{V}(x,y,z)\cdot d\vec{S}$

3. Volumen:

- $\int_V \vec{V}\, d\tau = \hat{x}\int V_x(\vec{r})\, d\tau + \hat{y}\int V_y(\vec{r})\, d\tau + \hat{z}\int V_z(\vec{r})\, d\tau$

Por otra parte,

$$\oint \vec{r}\cdot d\vec{r} = \oint (x,y,z)\cdot(dx,dy,dz) = \oint x\,dx + \oint y\,dy + \oint z\,dz$$

Con $\varphi(x,y,z) = \frac{1}{2}(x^2+y^2+z^2) \rightarrow \vec{F} = -\vec{\nabla}\varphi = -\left(\hat{x}\frac{\partial\varphi}{\partial x} + \hat{y}\frac{\partial\varphi}{\partial y} + \hat{z}\frac{\partial\varphi}{\partial z}\right) = -(x\hat{x} + y\hat{y} + z\hat{z}) = -\hat{r}$

Teorema de Gauss

$$\int_S \vec{V}\cdot d\vec{S} = \int_V \vec{\nabla}\cdot\vec{V}\, d\tau$$

$\rightarrow$ Teorema de Green $U(x,y,z)$ y $V(x,y,z)$ funciones escalares

$$\vec{\nabla}\cdot(U\vec{\nabla}V) = U\vec{\nabla}^2 V + (\vec{\nabla}U)(\vec{\nabla}V)$$
$$-\vec{\nabla}\cdot(V\vec{\nabla}U) = V\vec{\nabla}^2 U + (\vec{\nabla}V)(\vec{\nabla}U)$$

$$\vec{\nabla}\underbrace{(U\vec{\nabla}V - V\vec{\nabla}U)}_{\vec{V}} = \underbrace{U\vec{\nabla}^2 V - V\vec{\nabla}^2 U}_{\vec{\nabla}\vec{V}}$$

$\vec{\nabla}^2$: LAPLACIANO $= \vec{\nabla}\cdot\vec{\nabla}$

gradiente aplicado 2 veces

$\rightarrow \int_S (u\vec{\nabla}v - v\vec{\nabla}u)\, d\vec{S} = \int_V \vec{\nabla}\cdot\vec{V}\, d\tau$

Teorema de Stokes

$\vec{n}\,dA$

$$\oint_C \vec{V}\cdot d\vec{r} = \int_S (\vec{\nabla}\times\vec{V})\, d\vec{S}$$

Coordenadas Curvilíneas

$$\begin{aligned} x &= x(q_1,q_2,q_3) \\ y &= y(q_1,q_2,q_3) \\ z &= z(q_1,q_2,q_3) \end{aligned} \quad\longleftrightarrow\quad \begin{aligned} q_1 &= q_1(x,y,z) \\ q_2 &= q_2(x,y,z) \\ q_3 &= q_3(x,y,z) \end{aligned} \quad \in \mathbb{R}^3$$

ejemplo Coordenadas esféricas

$$\begin{aligned} x &= r\cos\varphi\sin\theta \\ y &= r\sin\varphi\sin\theta \\ z &= r\cos\theta \end{aligned} \quad\longleftrightarrow\quad \begin{aligned} r &= \sqrt{x^2+y^2+z^2} \\ \theta &= \cos^{-1}\left(\frac{z}{\sqrt{x^2+y^2+z^2}}\right) \\ \varphi &= \tan^{-1}\left(\frac{y}{x}\right) \end{aligned}$$

} fijar alguna produce una superficie

Pensamos en los unitarios $\hat{e}_i = \frac{1}{h_i}\frac{\partial\vec{r}}{\partial q_i}$, con $h_i = \left|\frac{\partial\vec{r}}{\partial q_i}\right|$ el factor de escala. Es decir, $\hat{e}_i = \frac{\frac{\partial\vec{r}}{\partial q_i}}{\left|\frac{\partial\vec{r}}{\partial q_i}\right|}$

$$\begin{aligned} \hat{e}_r &= \sin\theta\cos\varphi\,\hat{i} + \sin\theta\sin\varphi\,\hat{j} + \cos\theta\,\hat{k} \\ \hat{e}_\theta &= \cos\theta\cos\varphi\,\hat{i} + \cos\theta\sin\varphi\,\hat{j} - \sin\theta\,\hat{k} \\ \hat{e}_\varphi &= -\sin\varphi\,\hat{i} + \cos\varphi\,\hat{j} \end{aligned}$$

Ahora, $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

$$d\vec{S} = d\vec{r}^{\,2} = (dx)^2 + (dy)^2 + (dz)^2$$

$$d\vec{r} = \sum_{i=1}^{3}\frac{d\vec{r}}{dq_i}\,dq_i = \left(\frac{dx}{dr}\,dr + \frac{dx}{d\theta}\,d\theta + \frac{dx}{d\varphi}\,d\varphi\right)\hat{i} + \cdots$$

$$dV = dx\,dy\,dz$$

demostrar

$$dV = \begin{vmatrix} g_{11} & g_{12} & g_{13} \\ g_{21} & g_{22} & g_{23} \\ g_{31} & g_{32} & g_{33} \end{vmatrix}^{1/2} dq_1\,dq_2\,dq_3 = \sqrt{g}\,dq_1\,dq_2\,dq_3$$

si $dV = \left|\frac{d\vec{r}}{dq_1}\cdot\left(\frac{d\vec{r}}{dq_2}\times\frac{d\vec{r}}{dq_3}\right)\right|$

En general,

$g_{ij} = \sum_k \frac{\partial x_k}{\partial q_i}\frac{\partial x_k}{\partial q_j}$

$$d\vec{r}\cdot d\vec{r} = \sum_{i=1}^{3}\sum_{j=1}^{3} g_{ij}\,dq_i\,dq_j = \sum_{i=1}^{3}\sum_{j=1}^{3}\left(\sum_{k=1}^{3}\frac{\partial x_k}{\partial q_i}\frac{\partial x_k}{\partial q_j}\right)dq_i\,dq_j = dS^2$$

Si las coordenadas son ortogonales,

$g_{ij} = 0,\ i \neq j$

$\rightarrow ds^2 = \sum_{i=1}^{3} g_{ii}(dq_i)^2 = \sum_{i=1}^{3}(h_i\,dq_i)^2$ dado que $g_{ii} = h_i^2$

$\therefore d\vec{r} = \sum_{i=1}^{3} h_i\,dq_i\,\hat{e}_i = \sum_{i=1}^{3} ds_i\,\hat{e}_i$ → diferencial de línea

Operadores Diferenciales

$$\vec{\nabla}\psi(q_1,q_2,q_3) = \sum_{i=1}^{3}\hat{e}_i\frac{d}{ds_i}\psi(q) = \sum_{i=1}^{3}\frac{1}{h_i}\frac{d}{dq_i}\psi(q)$$ → GRADIENTE: $\vec{\nabla} = \sum_i \hat{e}_i\frac{1}{h_i}\frac{\partial}{\partial q_i}$

$$\vec{\nabla}\cdot\vec{V} = \frac{1}{\sqrt{g}}\sum_{i=1}^{3}\frac{\partial}{\partial q_i}\sqrt{\frac{g}{g_i}}\,V_i$$ → DIVERGENCIA

$$\vec{\nabla}^2\mathcal{C} = \frac{1}{\sqrt{g}}\sum_{i=1}^{3}\frac{\partial}{\partial q_i}\left(\frac{\sqrt{g}}{g_i}\frac{\partial}{\partial q_i}\mathcal{C}\right)$$ → LAPLACIANO

$$\vec{\nabla}\times\vec{V} = \frac{1}{\sqrt{g}}\begin{vmatrix} \sqrt{g_1}\,\hat{e}_1 & \sqrt{g_2}\,\hat{e}_2 & \sqrt{g_3}\,\hat{e}_3 \\ \frac{\partial}{\partial q_1} & \frac{\partial}{\partial q_2} & \frac{\partial}{\partial q_3} \\ \sqrt{g_1}\,V_1 & \sqrt{g_2}\,V_2 & \sqrt{g_3}\,V_3 \end{vmatrix}$$ → ROTACIONAL

Coordenadas Parabólicas

Se definen

(ξ, η, φ)

$$\begin{aligned} x &= \sqrt{\xi\eta}\cos\varphi \\ y &= \sqrt{\xi\eta}\sin\varphi \\ z &= \frac{1}{2}(\xi-\eta) \end{aligned} \quad \longleftrightarrow \quad \begin{aligned} \xi &= r+z \\ \eta &= r-z \\ \varphi &= \tan^{-1}\left(\frac{y}{x}\right) \end{aligned}$$

$$\left.\begin{aligned} \frac{\partial x}{\partial \xi} &= \frac{1}{2}\sqrt{\frac{\eta}{\xi}}\cos\varphi \\ \frac{\partial y}{\partial \xi} &= \frac{1}{2}\sqrt{\frac{\eta}{\xi}}\sin\varphi \\ \frac{\partial z}{\partial \xi} &= \frac{1}{2} \end{aligned}\right\} \quad h_\xi = \frac{1}{2}\sqrt{\frac{\eta}{\xi}+1}, \quad h_\eta = \sqrt{\frac{\xi}{\eta}+1}, \quad h_\varphi = \sqrt{\xi\eta} \rightarrow \text{ver cuaderno}$$

En general,

$$h_i^2 = g_{ii} = \frac{\partial \vec{r}}{\partial q_i}\frac{\partial \vec{r}}{\partial q_i} = \left(\frac{\partial \vec{r}}{\partial q_i}\right)^2$$

$$\rightarrow h_i = \sqrt{\left(\frac{dx}{dq_i}\right)^2+\left(\frac{dy}{dq_i}\right)^2+\left(\frac{dz}{dq_i}\right)^2} = \left|\frac{\partial \vec{r}}{\partial q_i}\right|$$

Por ejemplo, $dV = \sqrt{\frac{\xi+\eta}{4\xi}}\sqrt{\frac{\xi+\eta}{4\eta}}\sqrt{\xi\eta}\, d\xi\, d\eta\, d\varphi = \left(\frac{\xi+\eta}{4}\right)^2 d\xi\, d\eta\, d\varphi$

Espacios de Funciones

Definimos

$\{\varphi_n(x)\}_{n=1}^{\infty}$ el conjunto de funciones bien definidas en $x\in[a,b]$, como **BASE DE FUNCIONES**, tal que

$f(x) = \sum_{n=1}^{\infty} c_n \varphi_n(x)$, $c_n \in \mathbb{C}$

Si se cumple

$\sum_{n=1}^{\infty} c_n \varphi_n(x) = 0 \rightarrow c_n = 0$

La base $\{\varphi_n(x)\}_{n=1}^{\infty}$ será linealmente independiente

ejemplo. $\{x^n\}_{n=1}^{\infty}$, la base de los monomios con $x\in[a,b]$ $\rightarrow$ $\text{Pol} = \sum_{n=1}^{\infty} c_n x^n$

Producto Escalar

$$(f_1, f_2) = \int_a^b f_1^*(x) f_2(x)\, dx \quad \leftarrow \text{complejo conjugado}$$

$$(f, f) = \int_a^b f^*(x) f(x)\, dx = \int_a^b |f(x)|^2 dx$$

Si la integral converge para una función f, la función se denomina de cuadrado integrable

Si

$$(\varphi_n, \varphi_m) = \int_a^b \varphi_n^*(x)\, \varphi_m(x)\, dx = a_n \delta_{nm}$$

la base $\{\varphi_n\}_{n=1}^{\infty}$ se denomina **ORTOGONAL** Si, además, $a_n = 1$, se denomina **ORTONORMAL**

$$\tilde{\varphi}_n(x) = \frac{\varphi_n(x)}{\sqrt{a_n}} \rightarrow (\tilde{\varphi}_n(x), \tilde{\varphi}_n(x)) = \int_a^b |\tilde{\varphi}_n(x)|^2 dx = \rightarrow \text{completud}$$

- Propiedades

$$(\varphi_n, \varphi_m) = (\varphi_m, \varphi_n)^*$$

$$(a\varphi_n + b\varphi_m, \varphi_\ell) = a^*(\varphi_n, \varphi_\ell) + b^*(\varphi_m, \varphi_\ell)$$

$$(\varphi_\ell, a\varphi_n + b\varphi_m) = a(\varphi_\ell, \varphi_n) + b(\varphi_\ell, \varphi_m)$$

$$(\varphi_n, \varphi_n) = \int_a^b |\varphi_n(x)|^2 dx \geq 0$$

Ortogonalidad con Función de Peso

Si $p(x) \in [a,b]$

$$(\varphi_n, \varphi_m) = \int_a^b p(x)\, \varphi_n^*(x)\, \varphi_m(x)\, dx = a_n \delta_{nm}$$

Además, $\varphi_n^N(x) = \sqrt{\frac{p(x)}{a_n}}$ forma base ortonormal

Sea $\{\varphi_n(x)\}$ $x \in [a,b]$ base ortonormal.

$$f(x) = \sum_{n=1}^{\infty} c_n \varphi_n(x)$$

$$c_m = (\varphi_m, f) = \int_a^b \varphi_m^*(x) f(x)\, dx$$

$$\rightarrow f(x) = \sum_{n=1}^{\infty} \left(\int_a^b \varphi_n^*(x') f(x') dx' \right) \varphi_n(x) = \int_a^b dx' f(x') \underbrace{\left(\sum_{n=1}^{\infty} \varphi_n^*(x') \varphi_n(x) \right)}_{\delta(x-x') \rightarrow \text{Dirac}}$$

$$\rightarrow \delta(x-x') = \sum_{n=1}^{\infty} \underbrace{\varphi_n^*(x')}_{c_n} \varphi_n(x)$$

$$\int_0^b |f(x) - \sum_{k=1}^{n} c_k \varphi_k(x)|^2 dx < \varepsilon \rightarrow \text{mide la exactitud de la aproximación}$$

Ahora,

$$(f,g) = \left(\sum_n a_n \varphi_n, \sum_m b_m \varphi_m \right) = \sum_n \sum_m a_n^* b_m \underbrace{(\varphi_n, \varphi_m)}_{\delta_{nm}} = \sum_n a_n b_n$$

convolución

Para que la base esté normalizada,

$$(f,f) = \left(\sum_n a_n \varphi_n, \sum_m a_m \varphi_m \right) = \sum_n \sum_m a_n^* a_m (\varphi_n, \varphi_m) = \sum_n |a_n|^2 = 1$$

$$\rightarrow f^N(x) = \frac{1}{\sqrt{\sum_n |a_n|^2}} f(x)$$

En base no ortogonal

$$(\varphi_m, \varphi_n) = S_{nm} \neq \delta_{nm}$$

$$\varphi_n^N = \frac{1}{\sqrt{S_{nm}}} \varphi_n(x)$$

Completez

Sea $\{\varphi_n(x)\}_{n=1}^{\infty}$ una base ortonormal en $[a,b]$, y sea $f(x) = \sum_{n=1}^{\infty} c_n \varphi_n(x) \in [a,b]$. Entonces

$$c_m = (\varphi_m, f) = \int \varphi_m^*(x) f(x)\, dx, \text{ con } f(x) = \sum_{n=1}^{\infty} \left(\int \varphi_m^*(x') f(x')\, dx' \right) \varphi_n(x)$$

$$f(x) = \int dx' f(x') \left(\sum_{n=1}^{\infty} \varphi_n^*(x') \varphi_n(x) \right) \rightarrow \underbrace{\sum_{n=1}^{\infty} \varphi_n^*(x') \varphi_n(x) = \delta(x-x')}_{\text{RELACIÓN DE COMPLETEZ}}$$

Productos Internos

Con

$$f(x)=\sum_{n=1}^{\infty} a_n \varphi_n(x) \quad y \quad g(x)=\sum_{m=1}^{\infty} b_m \varphi_m(x)$$

1 $(f,f)=\left(\sum_{n=1}^{\infty} a_n\varphi_n(x), \sum_{m=1}^{\infty} a_m\varphi_m(x)\right)=\sum_{n=1}^{\infty} a_n^* \sum_{m=1}^{\infty} a_m (\varphi_n,\varphi_m)=\sum_{n,m} a_n^* a_m \delta_{nm} = \sum_n |a_n|^2$

2 $(g,g)=\sum_n |b_m|^2$

3 $(f,g)=\sum_n a_n^* \sum_m b_m (\varphi_n,\varphi_m)=\sum_n a_n^* b_n$

Así,

$$f^N(x)=\frac{1}{\sqrt{\sum_n |a_n|^2}} \rightarrow \text{Normalización de } f$$

$$\rightarrow (f^N,g^N)=\frac{1}{\sqrt{\sum |a_n|^2}}\frac{1}{\sqrt{\sum |b_n|^2}}\sum_n a_n^* b_n$$

Bases No-Ortogonales

Con $\{\varphi_n(x)\}_{n=1}^{\infty} \in [a,b]$

$$(\varphi_n,\varphi_m)=\int_a^b \varphi_n^* \varphi_m dx = S_{nm}, \text{ en general, } S_{nm}\neq S_{nm}$$

Podemos normalizar la base

$$\varphi_m^N(x)=\frac{1}{\sqrt{S_{mm}}}\varphi_m(x) \rightarrow (\varphi_m^N,\varphi_n^N)=\frac{1}{\sqrt{S_{mm}}}\frac{1}{\sqrt{S_{nn}}}(\varphi_m,\varphi_n)=\frac{S_{mn}}{\sqrt{S_{mm}S_{nn}}}, \text{ si } m=n \rightarrow (\varphi_m^N,\varphi_m^N)=\frac{S_{mm}}{\sqrt{S_{mm}S_{mm}}}=1$$

Así, $0\leq |S_{mn}|\leq 1$. Ahora,

$$(f,g)=\sum_n a_n^* \sum_m b_m (\varphi_n,\varphi_m)=\sum_n\sum_m a_n^* b_m S_{nm} \rightarrow \text{en general, } S_{nm}=S_{mn}^*, \text{ ya que la matriz métrica es simétrica}$$

Consideremos la desigualdad de Schwartz $|\vec{u}\cdot\vec{v}|\leq |\vec{u}||\vec{v}|$ En el espacio funcional

$$|(f,g)|^2\leq (f,f)(g,g)$$

$$\left|\sum_n a_n^* b_n\right|^2 \leq \left(\sum_n |a_n|^2\right)\left(\sum_m |b_m|^2\right)$$

Ortogonalización

Gram-Schmidt

$\{\vec{x}_1,\vec{x}_2,\cdots,\vec{x}_n\} \in \mathbb{R}^n$, LI no ortogonal

$$\rightarrow \begin{cases} \vec{v}_1=\vec{x}_1 \\ \vec{v}_2=\vec{x}_2-\dfrac{(\vec{x}_2\cdot\vec{v}_1)}{(\vec{v}_1,\vec{v}_1)}\vec{v}_1=\vec{x}_2-(\vec{x}_2\cdot\hat{v}_1)\hat{v}_1 \\ \vec{v}_3=\vec{x}_3-\dfrac{(\vec{x}_3\cdot\vec{v}_1)}{(\vec{v}_1,\vec{v}_1)}\vec{v}_1-\dfrac{(\vec{x}_3\cdot\vec{v}_2)}{(\vec{v}_2\cdot\vec{v}_2)}\vec{v}_2 = \vec{x}_3-(\vec{x}_3\cdot\hat{v}_1)\hat{v}_1-(\vec{x}_3\cdot\hat{v}_2)\hat{v}_2 \\ \vdots \\ \vec{v}_n=\vec{x}_n-\sum_{k=1}^{n-1}(\vec{x}_n,\hat{v}_k)\hat{v}_k \end{cases}$$

De la misma forma, para $\{f_0(x), f_1(x), \cdots, f_N(x)\} \rightarrow \{P_0^N(x), P_1^N(x), \cdots, P_N^N(x)\}$, con

$$P_n(x)=f_n(x)-\sum_{k=0}^{n-1}\frac{(f_n,P_k)}{(P_k,P_k)}P_k(x), \quad P_n^N(x)=\frac{f_n(x)-\sum_{k=0}^{n-1}\frac{(f_n,P_k)}{(P_k,P_k)}P_k(x)}{\left|f_n(x)-\sum_{k=0}^{n-1}\frac{(f_n,P_k)}{(P_k,P_k)}P_k(x)\right|}$$

Hagamos el proceso para $\{x^n\}$

$$\rightarrow \left\{\frac{1}{\sqrt{2}}, \sqrt{\frac{3}{2}}x, \sqrt{\frac{5}{2}}\left(\frac{3}{2}x^2-\frac{1}{2}\right), \cdots\right\} \rightarrow \text{POLINOMIOS DE LEGENDRE}$$

Ortogonalización Simétrica

$\{\phi_\mu(\vec{r})\}_{\mu=1}^{N}$ no ortogonal, pero normalizada $\rightarrow (\phi_\mu, \phi_\nu) = S_{\mu\nu}$, $0 \leq |S_{\mu\nu}| \leq 1$

$\{\phi'_\mu(\vec{r})\}_{\mu=1}^{N} \rightarrow (\phi'_\mu, \phi'_\nu) = \int d^3\vec{r}\ \phi'^{*}_\mu(\vec{r})\,\phi'_\nu(\vec{r}) = \delta_{\mu\nu}$

Pensemos en una transformación denotada por la matriz $\mathbb{X}$

$$\Phi' = \mathbb{X}\Phi \rightarrow \phi'_\mu = \sum_{\alpha=1}^{N} X_{\alpha\mu}\phi_\alpha, \quad \phi'_\nu = \sum_{\beta=1}^{N} X_{\beta\nu}\phi_\beta$$

$$\rightarrow (\phi'_\mu, \phi'_\nu) = \int d^3\vec{r} \left(\sum_\alpha X^{*}_{\alpha\mu}\phi'_\alpha\right)\left(\sum_\beta X_{\beta\nu}\phi_\beta\right) = \sum_\alpha\sum_\beta X^{\dagger}_{\alpha\mu} X_{\beta\nu} \underbrace{(\phi_\alpha, \phi_\beta)}_{S_{\alpha\beta}} = \sum_{\alpha\beta} X^{*}_{\alpha\mu} S_{\alpha\beta} X_{\beta\nu} \rightarrow \mathbb{1} = \mathbb{X}^{T}\mathbb{S}\mathbb{X}$$

Proponemos

$$\mathbb{X} = \mathbb{S}^{-1/2}$$

Notemos que $\mathbb{S}$ no es una matriz diagonal. Para evaluar la función, debemos diagonalizar la matriz. Para una matriz densa,

$$A = UaU^{\dagger} \rightarrow f(A) = Uf(a)U^{\dagger} \qquad \rightarrow U \text{ es la matriz de autovectores}$$

$$\therefore \mathbb{S}^{-1/2} = U\mathbb{S}^{-1/2}U^{\dagger}$$

Ortogonalización Canónica

$\mathbb{X} = U\mathbb{S}^{-1/2}$ → comprobar

La base Gaussiana $\phi_{\alpha_i}(x) = Ne^{-\alpha_i x^2}$, $\{\phi_{\alpha_i}\}_{i=1}^{\infty}$

$$S_{ij} = \langle \phi_{\alpha_i} | \phi_{\alpha_j} \rangle = |N_{\alpha_i}|^2 \int_{-\infty}^{\infty} dx\, e^{-\alpha_i x^2} e^{-\alpha_j x^2} = |N_{\alpha_i}|^2 \int e^{-(\alpha_i+\alpha_j)x^2} dx = N^2\sqrt{\frac{\pi}{2\alpha_i}}, \text{ tal que } N_i = \sqrt{\frac{2\alpha_i}{\pi}} \text{ si la base está normalizada}$$

Eligiendo

$$\phi^{n_i}_{\alpha_i}(x) = \{N(\alpha_i, n_i)\, x^{n_i} e^{-\alpha_i x^2}\}$$

la base será completa

Los Polinomios de Bersteiv son una base no ortonormal

$$B^n_m(x) = \binom{n}{m}\frac{1}{b-a^n}(x-a)^n(b-x)^{n-m}$$

La base compuesta

$$\left\{\cos\left(\frac{n\pi x}{2}\right), \sin\left(\frac{n\pi x}{2}\right)\right\}_{n=1}^{N}$$

es ortonormal. Entonces,

$$\delta(x-x') = \sum_{n=1}^{N} \cos\left(\frac{n\pi x}{2}\right)\cos\left(\frac{n\pi x'}{2}\right) + \sum_{n=1}^{N} \sin\left(\frac{n\pi y}{2}\right)\sin\left(\frac{n\pi x'}{2}\right)$$

es una representación válida de la Delta de Dirac

★ recordemos

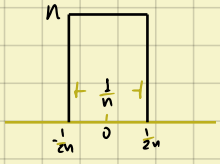
$$\binom{n}{m} = \frac{n!}{m!(n-m)!}$$

Representaciones de $\delta(x)$

$$\int_{-\infty}^{\infty} \delta(x)\,dx = 1$$

$$\int_{-\infty}^{\infty} f(x)\,\delta(x)\,dx = f(0)$$

1. $$\delta_n(x) = \begin{cases} 0, & x < -\frac{1}{2n} \\ n, & -\frac{1}{2n} \le x \le \frac{1}{2n} \\ 0, & x > \frac{1}{2n} \end{cases}$$



2. $$\delta_n(x) = \frac{n}{\sqrt{\pi}} e^{-n^2x^2}$$

3. $$\delta_n(x) = \frac{n}{\pi}\frac{1}{1+n^2x^2} \rightarrow \text{LORENTZIANA}\quad \mathcal{L}(x) = \frac{1}{\pi}\frac{\frac{\Gamma}{2}}{\left(\frac{\Gamma}{2}\right)^2 + (x-x_0)^2}$$
$$= \frac{1}{\pi}\frac{\frac{1}{n}}{\left(\frac{1}{n}\right)^2 + x^2}$$

4. $$\delta_n(x) = \frac{\sin(nx)}{\pi x}$$

Ejercicio

$$\int_{-\infty}^{\infty} f(x)\,\delta(x^2 - a^2)\,dx$$

Propiedades

1. Par: $\delta(x) = \delta(-x) \rightarrow \delta(x - x') = \delta(x' - x)$

2. $\delta(ax) = \frac{1}{|a|}\delta(x)$, $a \in \mathbb{C}$

3. $$\int_{-\infty}^{\infty} \delta(x-a)\,\delta(x-b)\,dx = \delta(a-b) = \delta(b-a)$$

4. $\delta(g(x)) \rightarrow g(x) \approx g(a_n) + g'(a_n)(x - a_n) + \dots$

$$\rightarrow \int f(x)\,\delta(g(x))\,dx = \sum_n \int f(x)\,\delta(g'(a_n)(x-a_n))\,dx = \sum_n \int f(x)\frac{1}{|g'(a_n)|}\delta(x-a_n)\,dx = \sum_n \frac{1}{|g'(a_n)|} f(a_n)$$

Derivada de $\delta(x)$

Ya que $\delta(x)$ es par, $\frac{d\,\delta(x)}{dx}$ es impar $\rightarrow \delta'(x) = -\delta'(-x)$. En general

$$\frac{d^n\delta(x)}{dx^n} \begin{cases} \text{par si } n \text{ par} \\ \text{impar si } n \text{ impar} \end{cases}$$

Veamos

$$\int \underbrace{f(x)}_{u}\,\underbrace{\frac{d}{dx}\delta(x)\,dx}_{dv} = \cancel{f(x)\delta(x)}^{\delta(\pm\infty)=0} - \int \delta(x) f'(x)\,dx = -f'(0)$$

Con $f(x)$ impar, $-f_p'(0) = f_I(0) = 0$

$$\rightarrow \int f(x)\frac{d}{dx}\delta(x-x')\,dx = -f'(x')$$

Expansión

Con $\{\varphi_n(x)\}_{n=1}^{\infty}$

$$\delta(x-t) = \sum_{n=1}^{\infty} c_n \varphi_n(x), \quad c_m = (\varphi_m, \delta) = \int_{-\infty}^{\infty} dx'\,\varphi_m(x')\,\delta(x'-t) = \varphi_m(t)$$
$$= \sum_{n=1}^{\infty} \varphi_n(t)\,\varphi_n(x) \rightarrow \text{para cualquier base bien comportada}$$

Tal que

$$F(x) = \int F(t)\,\delta(t-x)\,dt = \int \left(\sum_k a_k \varphi_k(t)\right)\left(\sum_n \varphi_n(t)\,\varphi_n(x)\right) dt = \sum_k \sum_n a_k \varphi_n(x) \int \varphi_k(t)\,\varphi_n(t)\,dt = \sum_k \sum_n a_k \varphi_n(x)\,\delta_{kn} = \sum_n a_n \varphi_n(x)$$

Funciones de Base de Índice Continuo

$$F(x) = \int \underbrace{dk\, G(k)}_{c_i} \underbrace{\frac{1}{\sqrt{2\pi}} e^{ikx}}_{\varphi_i(x)}, \quad G(k) = \int \underbrace{dx\, F(x)}_{c_i} \underbrace{\frac{1}{\sqrt{2\pi}} e^{-ikx}}_{\varphi_i(x)} \rightarrow \text{TRANSFORMADA DE FOURIER}$$

Análogo a lo anterior, podemos expandir funciones en una base continua. En el caso particular de $\frac{1}{\sqrt{2\pi}} e^{ikx}$,

$(v_k, v_k) = \frac{1}{2\pi} \int e^{ikx} e^{-ikx} dx \rightarrow$ diverge, es decir, $\frac{1}{\sqrt{2\pi}} e^{ikx} \notin L^2$

Veamos, $\delta(x-x') = \sum_i u_i^*(x)\, u_i(x') \rightarrow \int dk\, v_k^*(x)\, v_k(x') = \frac{1}{2\pi} \int dk\, e^{ik} e^{-ikx'} = \frac{1}{2\pi} \int dk\, e^{-ik(x-x')} \overset{x'=0}{=} \frac{1}{2\pi} \int_{-\infty}^{\infty} dk\, e^{ikx}$ → Representación exacta de $\delta(x)$

Ahora,

$$\psi(x) = \int \varphi(k) \frac{1}{\sqrt{2\pi}} e^{ikx} dk, \quad \varphi(k) = e^{-\frac{(k-k_0)^2}{\gamma^2}}$$ → γ es la anchura cuando la gaussiana cae a $\frac{1}{e}$

$$= \frac{1}{\sqrt{2\pi}} \int e^{-\frac{(k-k_0)^2}{\gamma^2}} e^{ikx} dk \rightarrow \text{completar cuadrados en el exponente}$$

$$= \frac{1}{\sqrt{2}} \gamma e^{ik_0 x} e^{-\gamma^2 \frac{x^2}{4}}$$

Generalización de δ

En $\mathbb{R}^3$

$$\delta(\vec{r}-\vec{r}') = \delta(x-x')\,\delta(y-y')\,\delta(z-z')$$

En general,

$$\delta(\vec{r}-\vec{r}') = \frac{1}{h}\,\delta(q_1-q_1')\,\delta(q_2-q_2')\,\delta(q_3-q_3')$$

- esféricas: $\frac{1}{r^2 \sin\theta}\,\delta(r-r')\,\delta(\theta-\theta')\,\delta(\varphi-\varphi') \rightarrow \frac{1}{2\pi r^2 \sin\theta}\,\delta(r-r')\,\delta(\theta-\theta')$ a 2 dim si hay simetría (se hace $\int h\, d\varphi$)
- parabólicas: $\frac{4}{\xi+\eta}\,\delta(\xi-\xi')\,\delta(\eta-\eta')\,\delta(\varphi-\varphi')$

2. Teoría de Sturm-Liouville

Ecuaciones Diferenciales Parciales Lineales

$\mathcal{L}[u(x,y)] = G(x,y)$: $\mathcal{L}$ es un super-operador; representa la EDPL

↳ $F(u_{xx}, u_{yy}, u_{xy}, u_x, u_y, u(x,y), x, y) = 0$

$Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = G$ → EDPL

Principio de superposición

$$\left.\begin{aligned} \mathcal{L}[u_1(x,y)] &= G \\ \mathcal{L}[u_2(x,y)] &= G \end{aligned}\right\} \mathcal{L}[c_1 u_1(x,y) + c_2 u_2(x,y)] = G$$

Construimos una ecuación característica

$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ → se puede rotar hacia otro sistema de referencia

En el sistema rotado,

$A'x'^2 + C'y'^2 + D'x' + E'y' + F' = 0$

$\rightarrow A'(x'+a)^2 + C'(y'+b)^2 + d = 0$

Ejemplos Ecuaciones de

- Laplace: $\nabla^2 \phi = 0$
- Poisson: $\nabla^2 \phi = -\frac{\rho}{\epsilon_0}$
- Helmholtz: $\nabla^2 \phi \pm k^2 \phi = 0$
- Onda: $\nabla^2 \phi - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \phi = 0$
- Difusión: $\nabla^2 \phi - \frac{1}{a^2} \frac{\partial}{\partial t} \phi = 0$
- Schrödinger:

TDSE → $\left(-\frac{\hbar^2}{2m}\nabla^2 + V - i\hbar\frac{\partial}{\partial t}\right)\phi = 0$

TISE → $\left(-\frac{\hbar^2}{2m}\nabla^2 + V - E\right)\phi = 0$

Tal que

$$A' = A'$$
$$C' = C'$$
$$D' = 2A'a$$
$$E' = 2C'b$$
$$F' = A'a^2 + C'b^2 + d$$

$$\rightarrow a = \frac{D'}{2A'}, \quad b = \frac{E'}{2C'}, \quad d = F' - \frac{D'^2}{4A'} - \frac{E'^2}{4C'}$$

$$\rightarrow A'(x'-x_0)^2 + C'(y'-y_0)^2 = \frac{D'^2}{4A'} + \frac{E'^2}{4C'} - F$$

- $A'C' < 0$ → hipérbola
- $A'C' > 0$ → elipse
- $A'C' = 0$ → parabola

De forma análoga, transformamos las derivadas

$$\begin{pmatrix} u_x \\ u_y \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} u'_x \\ u'_y \end{pmatrix} \quad \text{y} \quad \begin{pmatrix} u_{xx} \\ u_{xy} \\ u_{yy} \end{pmatrix} = \begin{pmatrix} \cos^2\theta & -2\sin\theta\cos\theta & \sin^2\theta \\ \sin^2\theta & 2\sin\theta\cos\theta & \cos^2\theta \\ \sin^2\theta & 2\sin\theta\cos\theta & \cos^2\theta \end{pmatrix} \begin{pmatrix} u'_{xx} \\ u'_{xy} \\ u'_{yy} \end{pmatrix}$$

¡Hacer u_{xy} con la definición!

$$\rightarrow \underbrace{u'_{xx}(A\cos^2\theta + B\sin\theta\cos\theta + C\sin^2\theta)}_{A'} + u'_{xy}\underbrace{(-A\sin 2\theta + B\cos 2\theta + C\sin 2\theta)}_{B'} + u'_{yy}\underbrace{(A\sin^2\theta - B\sin\theta\cos\theta + C\cos^2\theta)}_{C'} + u'_x\underbrace{(D\cos\theta + E\sin\theta)}_{D'} + u'_y\underbrace{(-D\sin\theta + E\cos\theta)}_{E'} + Fu = G$$

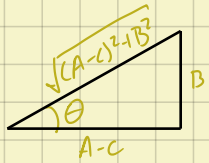
De B', $(C-A)\sin 2\theta + B\cos 2\theta = 0$ si queremos estructura cónica, entonces $\tan(2\theta) = \frac{B}{A-C}$

$$\rightarrow A'\left(\frac{\partial}{\partial x'} + \frac{D'}{2A'}\right)^2 u' + C'\left(\frac{\partial}{\partial y'} + \frac{E'}{2C'}\right)^2 u' + F'u' = G', \quad F' = F - \frac{D'^2}{4A'} - \frac{E'^2}{4C'}$$

→ Se cumple específicamente en el ángulo $\theta = \frac{1}{2}\tan^{-1}\left(\frac{B}{A-C}\right)$

- $A'C' < 0$ → hipérbola
- $A'C' > 0$ → elipse
- $A'C' = 0$ → parabola

Ahora, tenemos que expresar en términos del ángulo doble

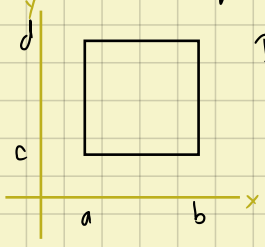


$A' \cdot C'$ se welve

$$\frac{4AC - B^2}{4} \rightarrow \begin{cases} 4AC - B^2 > 0, & \text{elíptica} \\ 4AC - B^2 < 0, & \text{hiperbólica} \\ 4AC - B^2 = 0, & \text{parabólica} \end{cases}$$

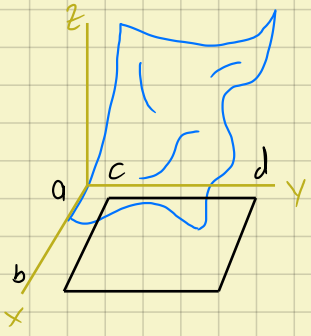
Condiciones de Contorno

1. Dirichlet: $u(x,y)$ definida en todos los puntos del contorno

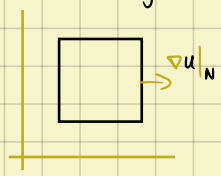


Debemos conocer el valor de u en cada punto del contorno

$$u(a,c) = f(y), \quad u(x,a) = h(x)$$
$$u(b,d) = g(y), \quad u(x,b) = i(x)$$



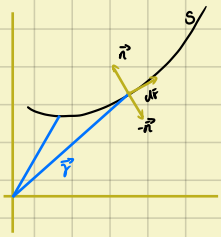
2. Neumann: el gradiente, en su componente normal, está definido en el contorno



3 Cauchy: $u(x,y)$ y $\nabla u|_N$ están bien definidos

¿Dónde se aplican?

Elípticas	$A'C' > 0$	Dirichlet o Neumann
Hiperbólicas	$A'C' < 0$	Cauchy
Parabólicas	$A'C' = 0$	Dirichlet o Neumann



$\vec{r} = (x,y)$

$\vec{r}(s) = (x(s), y(s))$

$d\vec{r}(s) = (dx(s), dy(s))$

$d\vec{r} \cdot \vec{n}\, ds = (dx(s), dy(s))(dy(s), -dx(s)) = 0$ (perpendiculares)

$\frac{du}{ds} = \nabla_i u \frac{d\vec{r}}{ds} = \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}\right)\left(\frac{\partial x}{\partial s}, \frac{\partial y}{\partial s}\right)$ → derivada tangencial

$\vec{n}\, ds(s) = (dy(s), -dx(s))$

$\vec{n} = \left(\frac{dy}{ds}, \frac{dx}{ds}\right)$

$\frac{du}{dn} = \nabla_i u \cdot \vec{n} = \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}\right)\left(\frac{dy}{ds}, \frac{dx}{ds}\right)$ → derivada normal

En general,

$$\left.\begin{aligned} \frac{d}{ds} &= \frac{dx}{ds}\frac{d}{dx} + \frac{dy}{ds}\frac{d}{dy} \\ \frac{d}{dn} &= \frac{dy}{ds}\frac{d}{dx} + \frac{dx}{ds}\frac{d}{dy} \end{aligned}\right\} \text{sistema de ecuaciones} \rightarrow \begin{pmatrix} \frac{dx}{ds} & \frac{dy}{ds} \\ \frac{dy}{ds} & -\frac{dx}{ds} \end{pmatrix}\begin{pmatrix} u_x \\ u_y \end{pmatrix} = \begin{pmatrix} \frac{du}{ds} \\ \frac{du}{dn} \end{pmatrix}$$

$A \cdot X = B$, $A^{-1} = \frac{1}{\det A}\tilde{A}^T$, $\tilde{A}^T = \begin{pmatrix} -\frac{dx}{ds} & -\frac{dy}{ds} \\ -\frac{dy}{ds} & \frac{dx}{ds} \end{pmatrix}$

$\det(A) = -\left(\frac{dx}{ds}\right)^2 - \left(\frac{dy}{ds}\right)^2 \neq 0$ → condición

Así,

$$\begin{vmatrix} \frac{dx}{ds} & \frac{dy}{ds} \\ \frac{dy}{ds} & -\frac{dx}{ds} \end{vmatrix} \neq 0$$

En un orden superior

$$\frac{d}{ds}(u_x) = \frac{dx}{ds}u_{xx} + \frac{dy}{ds}u_{xy}, \quad \frac{d}{ds}(u_y) = \frac{dx}{ds}u_{xy} + \frac{dy}{ds}u_{yy}$$

Además, tenemos $Au_{xx} + Bu_{xy} + Cu_{yy} = G$

$$\begin{pmatrix} \frac{dx}{ds} & \frac{dy}{ds} & 0 \\ 0 & \frac{dx}{ds} & \frac{dy}{ds} \\ A & B & C \end{pmatrix}\begin{pmatrix} u_{xx} \\ u_{xy} \\ u_{yy} \end{pmatrix} = \begin{pmatrix} \frac{du_x}{ds} \\ \frac{du_y}{ds} \\ G \end{pmatrix} \rightarrow \begin{vmatrix} \frac{dx}{ds} & \frac{dy}{ds} & 0 \\ 0 & \frac{dx}{ds} & \frac{dy}{ds} \\ A & B & C \end{vmatrix} = 0$$

$\rightarrow \frac{dx}{ds}\left(C\frac{dx}{ds} - B\frac{dy}{ds}\right) + A\left(\frac{dy}{ds}\right)^2 = 0 \rightarrow \boxed{A\left(\frac{dy}{ds}\right)^2 - B\frac{dx}{ds}\frac{dy}{ds} + C\left(\frac{dx}{ds}\right)^2 = 0}$ *2 direcciones características ortogonales

$\rightarrow A(dy)^2 - B\,dx\,dy + C(dx)^2 = 0 \quad \div (ds)^2$ ECUACIÓN CARACTERÍSTICA

$A - B\frac{dx}{dy} + C\left(\frac{dx}{dy}\right)^2 = 0 \quad \div (dy)$

$\therefore \frac{dx}{dy} = \frac{B \pm \sqrt{B^2 - 4AC}}{2C}$

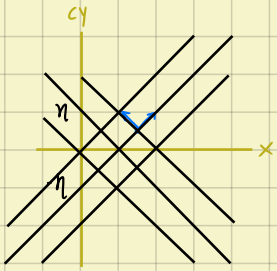
ejemplo $\frac{d^2}{dx^2}\phi - \frac{1}{c^2}\frac{d^2}{dt^2}\phi = 0$

$EC: \left(\frac{dy}{ds}\right)^2 - \frac{1}{c^2}\left(\frac{dx}{ds}\right)^2 = 0$

$(dy)^2 - \frac{1}{c^2}(dx)^2 = 0$

$(dy)^2 = \frac{1}{c^2}(dx)^2$

$\rightarrow \left(\frac{dx}{dy}\right)^2 = c^2 \rightarrow \frac{dx}{dy} = \pm c \rightarrow x = \eta \pm cy \rightarrow$



Un contorno en estas direcciones NO satisface las segundas derivadas de la ecuación. Es decir, las EC definen dos direcciones donde la EDP no tiene solución; esta solo se da en contornos con puntos no tangenciales a las EC

EDPL con coeficientes constantes

$Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = 0$

Suponemos $u = ce^{mx}e^{ny} = ce^{mx+ny}$

$u_x = mu \rightarrow u_{xx} = m^2u$

$u_y = nu \rightarrow u_{yy} = n^2u$

Rotamos para quitar Bu_{xy},

$Am^2u + Cn^2u + Dmu + Enu + Fu = 0$

$\rightarrow u(Am^2 + Cn^2 + Dm + En + F) = 0$, con $u \neq 0$

ejemplo $u_{xx} - u_{yy} - 2u_x + u = 0$

E.C. $m^2 - n^2 - 2m + 1 = 0$

$(m-1)^2 - n^2 = 0$

$n^2 = (m-1)^2 \rightarrow n = \pm(m-1)$

$\therefore u(x,y) = Ce^{mx}\left(c_1e^{(m-1)y} + c_2e^{-(m-1)y}\right)$

ejemplo $u_{xx} - \frac{1}{v^2}u_{yy} = 0$

EC $m^2 - \frac{1}{v^2}n^2 = 0$

$n = \pm vm$

$\therefore u(x,y) = e^{mx}\left(C_1e^{vmy} + C_2e^{-vmy}\right)$

ejemplo

Proponemos $u_{xx} - u_y = 0$

$u(x,y,\cdots) = A(x)B(y)\cdots$

$u_x = \frac{dA}{dx}B, \quad u_{xx} = \frac{d^2A}{dx^2}B$

$u_y = A\frac{dB}{dy}$

$\rightarrow \frac{d^2A}{dx^2}B - A\frac{dB}{dy} = 0 \rightarrow \frac{d^2A}{dx^2} = \frac{A}{B}\frac{dB}{dy} \rightarrow \frac{1}{A}\frac{d^2A}{dx^2} = \frac{1}{B}\frac{dB}{dy} = \alpha^2$ (constante)

$\begin{cases} \frac{d^2A}{dx^2} - \alpha^2A = 0 \\ \frac{dB}{dy} - \alpha^2B = 0 \end{cases} \rightarrow \begin{cases} A(x) = C_1e^{\alpha x} + C_2e^{-\alpha x} \\ B(y) = C_3e^{\alpha^2 y} \end{cases} \rightarrow u(x,y) = e^{\alpha^2 y}\left(C_1e^{\alpha x} + C_2e^{-\alpha x}\right)$

Debemos considerar

$$\begin{cases} \alpha^2, & u(x,y) = (c_1 e^{\alpha x} + c_2 e^{-\alpha x}) e^{\alpha^2 y} \\ -\beta^2, & u(x,y) = (d_1 e^{i\beta x} + d_2 e^{-i\beta x}) e^{-\beta^2 y} \\ 0, & u(x,y) = ax + b \end{cases}$$

que son todos los valores posibles para la constante de separación. En general,

$$u(x,y) = \int_0^\infty (c_1 e^{\alpha x} + c_2 e^{-\alpha x}) e^{\alpha^2 y} \, d\alpha + \int_0^\infty (d_1 e^{i\beta x} + d_2 e^{-i\beta x}) e^{-\beta^2 y} \, d\beta + ax + b$$

Separación de Variables

En coordenadas cartesianas 2D,

$$Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = 0$$

Probamos $u(x,y) = X(x)Y(y)$

$$\frac{A\ddot{X}Y}{XY} + \frac{B\dot{X}\dot{Y}}{XY} + \frac{CX\ddot{Y}}{XY} + \frac{D\dot{X}Y}{XY} + \frac{EX\dot{Y}}{XY} + \frac{FXY}{XY} = 0$$

$$\left(\frac{A\ddot{X}}{X} + \frac{D\dot{X}}{X}\right) + B\frac{\dot{X}\dot{Y}}{XY} + \left(\frac{C\ddot{Y}}{Y} + \frac{E\dot{Y}}{Y}\right) + F = 0$$

La ecuación es separable si

1. $A = A(x)$, $D = D(x)$
2. $B = 0$
3. $C = C(y)$, $E = E(y)$
4. $F = F(x)$ o $F = F(y)$ o $F = cte$

Asumiendo lo anterior,

$$\left(\frac{A\ddot{X}}{X} + \frac{D\dot{X}}{X} + \overbrace{F}^{f(x)}\right) = -\left(\frac{C\ddot{Y}}{Y} + \frac{E\dot{Y}}{Y}\right) = \pm\alpha^2$$

$$\begin{cases} A\ddot{X} + D\dot{X} + FX = \pm\alpha^2 X \\ C\ddot{Y} + E\dot{Y} = \mp\alpha^2 Y \end{cases} \rightarrow \begin{cases} X(x;\alpha) \\ Y(y;\alpha) \end{cases}$$

ejemplo eq de Laplace en 2D

$\nabla^2 u = 0 \rightarrow u_{xx} + u_{yy} = 0$, proponemos $u = A(x)B(y)$

$$\frac{\ddot{A}B}{u} + \frac{A\ddot{B}}{u} = \frac{0}{u}$$

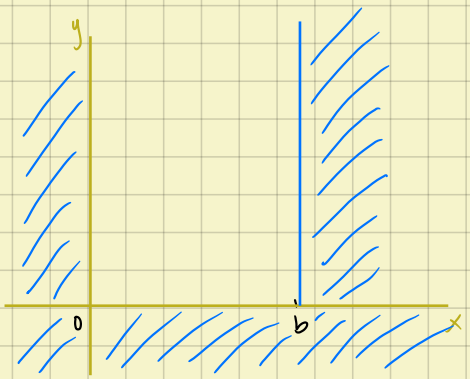
$$\frac{\ddot{A}}{A} + \frac{\ddot{B}}{B} = 0 \rightarrow \frac{\ddot{A}}{A} = -\frac{\ddot{B}}{B} = -\alpha^2$$

$$\begin{cases} \ddot{A} + \alpha^2 A = 0 \\ \ddot{B} - \alpha^2 B = 0 \end{cases} \rightarrow \begin{cases} A(x) = c_1 e^{i\alpha x} + c_2 e^{-i\alpha x} \\ B(y) = d_2 e^{\alpha y} + d_2 e^{-\alpha y} \end{cases}$$

Si $\alpha = 0$,

$\ddot{A} = 0 \rightarrow \dot{A} = a \rightarrow A = ax + b$ y $B = cy + d$

$$\rightarrow u(x,y) = \int (A(\alpha)e^{i\alpha x} + B(\alpha)e^{-i\alpha x})(C(\alpha)e^{\alpha x} + D(\alpha)e^{-\alpha x}) \, d\alpha + (ax+b)(cy+d)$$



$0 \le x \le b$
$0 \le y < \infty$; Consideramos

$$\left.\begin{aligned} u(x,0) &= f(x) \\ u(0,y) &= 0 \\ u(b,y) &= 0 \\ u(x,\infty) &= 0 \end{aligned}\right\} \text{Dirichlet}$$

Aplicamos las condiciones de frontera

$$\to u(x,y) = \int_0^\infty \left(A(\alpha)e^{i\alpha x} + B(\alpha)e^{-i\alpha x}\right)\left(C(\alpha)e^{\alpha x} + D(\alpha)e^{-\alpha x}\right) d\alpha \quad + (ax+b)(cy+d)$$

$$\to u(x,y) = \int_0^\infty d\alpha\, 2i A(\alpha) \sin(\alpha x) e^{-\alpha y}$$

$$= \int_0^\infty d\alpha \sum_n C_n \delta\left(\alpha - \frac{n\pi}{L}\right) \sin(\alpha x) e^{-\alpha y} = \sum_n C_n \sin\left(\frac{n\pi x}{L}\right) e^{\frac{n\pi y}{L}}$$

$$= \sum_n \left(\int_0^L \sin\left(\frac{n\pi x'}{L}\right) f(x')\,dx'\right) \sin\left(\frac{n\pi x}{L}\right) e^{-\frac{n\pi y}{L}}$$

Ejercicios

1. $u_{xx} + u_{yy} = 0$, $x \in [0,a]$, $y \in [0,b]$

$u(0,y) = 0$
$u(a,y) = 0$
$u(x,0) = 0$
$u(x,b) = V(x)$

2. $u_{xx} + u_{yy} - 2u_y + u = 0$, $x \in [0,L]$, $y \in [0,L]$

$u(0,y) = 0$
$u(L,y) = 0$
$u(x,0) = V(x)$
$u(x,L) = 0$

Separación de Variables 3D

$(\vec{\nabla}^2 + k^2)\psi = 0$, sea $\psi = X(x)Y(y)Z(z)$

$$\to YZ\frac{d^2X}{dx^2} + XZ\frac{d^2Y}{dy^2} + XY\frac{d^2Z}{dz^2} + k^2XYZ = 0$$

$$\frac{X''}{X} + \frac{Y''}{Y} + \frac{Z''}{Z} + k^2 = 0 \to \underbrace{\frac{X''}{X}}_{f(x)} = \underbrace{-k^2 - \frac{Y''}{Y} - \frac{Z''}{Z}}_{g(y,z)} = -l^2$$

1. $X'' = -l^2 X$

2. $-k^2 - \frac{Y''}{Y} - \frac{Z''}{Z} = -l^2 \to \underbrace{\frac{Y''}{Y}}_{h(y)} = \underbrace{-k^2 + l^2 - \frac{Z''}{Z}}_{i(z)} = -m^2$

2.1 $Y'' = -m^2 Y$

2.2 $Z'' = Z(\underbrace{l^2 + m^2 - k^2}_{-n^2})$

$Z'' = -Zn^2$

Coordenadas Cilíndricas

$\vec{\nabla}^2\psi = 0$, sea $\psi = R(\rho)\Phi(\varphi)Z(z)$

$$\left[\frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial}{\partial\rho}\right) + \frac{1}{\rho^2}\frac{\partial^2}{\partial\varphi^2} + \frac{\partial^2}{\partial z^2}\right]\psi = 0$$

$$\Phi Z\frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial}{\partial\rho}R\right) + RZ\frac{1}{\rho^2}\frac{\partial^2}{\partial\varphi^2}\Phi + R\Phi\frac{\partial^2}{\partial z^2}Z = 0 \quad \times \frac{\rho^2}{R\Phi Z} \to \psi$$

potencia más alta de una variable

$$\frac{\rho}{R}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial}{\partial\rho}R\right) + \frac{1}{\Phi}\frac{\partial^2}{\partial\varphi^2}\Phi + \frac{\rho^2}{Z}\frac{\partial^2}{\partial z^2}Z = 0$$

$$\frac{\Phi''}{\Phi} = -\frac{\rho^2}{Z}Z'' - \frac{\rho}{R}\frac{d}{d\rho}(\rho R') = -\nu^2$$

$\nu = \pm im$
$\nu^2 = (i^2)m^2$
$\nu^2 = -m^2$

1. $\Phi'' + \nu^2\Phi = 0 \to \Phi = C_1\cos(\nu\varphi) + C_2\sin(\nu\varphi)$,

2. $\frac{1}{Z}Z'' + \frac{1}{\rho R}\frac{d}{d\rho}(\rho R') = m^2 \to \frac{Z''}{Z} = -\frac{1}{\rho R}\frac{d}{d\rho}(\rho R') = -k^2$

2.1 $Z'' = -k^2 Z \to Z = d_1 e^{ikz} + d_2 e^{-ikz}$

2.2 $\frac{1}{\rho}\frac{d}{d\rho}(\rho R') + \left(k^2 - \frac{m^2}{\rho^2}\right)R = 0$ hacer $x = k\rho$

Coordenadas Esféricas

$\nabla^2 \psi = 0$, sea $\psi = \frac{U(r)}{r} y(\theta, \varphi)$

$$\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right] Uy = 0$$

$$y\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)\frac{U}{r} + \frac{U}{r}\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) y + \frac{U}{r}\frac{1}{r^2\sin^2\theta}\frac{\partial^2 y}{\partial\varphi^2} = 0$$

$$y\frac{1}{r}U'' + \frac{U}{r^3\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) y + \frac{U}{r^3\sin^2\theta} y'' = 0 \quad \cdot \frac{r^3}{Uy}$$

$$\frac{r^2}{U}U'' = -\frac{1}{y\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) y - \frac{1}{y\sin^2\theta} y'' = -b$$

1. $r^2U'' + bU = 0$
2. $\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) y + \frac{1}{\sin^2\theta}\frac{\partial^2 y}{\partial\varphi^2} + by = 0$

Nota: toda EDP asociada a un potencial central es separable en coordenadas esféricas

Ecuaciones de Autovalores

$H_{N\times N}\, \chi = \lambda\chi \quad , \quad \lambda \in \mathbb{C}$

$$\begin{pmatrix} H_{11} & H_{12} & \cdots & H_{1N} \\ H_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ H_{N1} & \cdots & & H_{NN} \end{pmatrix}\begin{pmatrix} \chi_1^1 & \chi_1^2 & \cdots & \chi_1^N \\ \chi_2^1 & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \chi_N^1 & \cdots & & \chi_N^N \end{pmatrix} = \begin{pmatrix} \lambda_1 & & & \emptyset \\ & \lambda_2 & & \\ & & \ddots & \\ \emptyset & & & \lambda_N \end{pmatrix}\begin{pmatrix} \chi_1^1 & \chi_1^2 & \cdots & \chi_1^N \\ \chi_2^1 & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \chi_N^1 & \cdots & & \chi_N^N \end{pmatrix}$$

$H\chi_j = \lambda_j \chi_j \rightarrow \chi_j^+ H^+ = \chi_j^+ \lambda_j^*$

$\chi_j^+ H^+ \chi_i = \lambda_i \chi_j^+ \chi_i$

$-(\chi_j^+ H \chi_i = \lambda_j^* \chi_j^+ \chi_i)$

$0 = (\lambda_i - \lambda_j^*)\chi_j^+ \chi_i$

$\rightarrow \lambda_i = \lambda_j^* \rightarrow \lambda_i \in \mathbb{R}$ ó $\chi_j^+ \chi_i = 0 \rightarrow \chi_i \perp \chi_j$

Operador Lineal de Orden 2

$$\mathcal{L}[f(x)] = a_2(x)D^2 f(x) + a_1(x)Df(x) + a_0(x)f(x) = g(x) \quad , \quad \{f(x), a_2, a_1, a_0\} \in \mathbb{C}$$

$\mathcal{L}[\,] = a_2 D^2[\,] + a_1 D[\,] + a_0[\,]$ → sobre una función

Así,

$\mathcal{L}[f(x)] = g(x)$

es una ecuación lineal de 2do orden

Ahora, veamos el producto escalar en $[a,b]$

$$(\mathcal{L}f, g) = \int_a^b (\mathcal{L}f)^* g\, dx = \int_a^b (a_2^* D^2 f^* + a_1^* Df^* + a_0^* f^*) g\, dx = \int_a^b (\underbrace{ga_2^* D^2 f^*}_{A} + \underbrace{ga_1^* Df^*}_{B} + ga_0^* f^*)\, dx$$

$AD^2 f^* = D[ADf^* - f^* DA] + f^* D^2 A = DADf^* + AD^2 f^* - Df^* DA - f^* D^2 A + f^* D^2 A$

$BDf^* = D[Bf^*] - DBf^*$

entonces,

$$(\mathcal{L}f, g) = \int_a^b \left\{ D[ga_2^* Df^* - f^* D(ga_2^*)] + f^* D^2(ga_2^*) + D(ga_1^* f^*) - D(ga_1^*) f^* + ga_0^* f^* \right\} dx$$

$$= \int_a^b f^* \left\{ D^2[a_2^* g] - D[a_1^* g] + a_0^*[g] \right\} dx$$

$$+ \left\{ g a_2^* Df^* - f^* D[ga_2^*] + ga_1^* f^* \right\}_a^b$$

podemos definir

$$\bar{\mathcal{L}}[\cdot] = D^2(a_2^*[\cdot]) - D(a_1^*[\cdot]) + a_0^*[\cdot]$$

entonces,

$$(\mathcal{L}f(x), g(x)) = \int_a^b f^*(x) \bar{\mathcal{L}}[g(x)] dx = (f, \bar{\mathcal{L}}g) + \left\{ g a_2^* Df^* - f^* D[ga_2^*] + ga_1^* f^* \right\}_a^b$$

$$= (f, \bar{\mathcal{L}}g) + \left\{ a_2^*(g\dot{f}^*) - f^*(\dot{a}_2^* g + a_2^* \dot{g}) + a_1^* g f^* \right\}_a^b = (f, \bar{\mathcal{L}}g) + \left\{ a_2^*(g\dot{f}^* - f^*\dot{g}) + f^* g [a_1^* - \dot{a}_2^*] \right\}_a^b$$

notemos que

$$W(f^*, g) = \begin{vmatrix} f^* & g \\ \dot{f}^* & \dot{g} \end{vmatrix} = f^*\dot{g} - g\dot{f}^*$$

entonces,

$$(\mathcal{L}f, g) = (f, \bar{\mathcal{L}}g) + \left\{ a_2^* W(f^*, g) + f^* g (a_1^* - \dot{a}_2^*) \right\}_a^b$$

OPERADOR AUTOADJUNTO

Se dice autoadjunto a un operador $\mathcal{L}$ si

$$\mathcal{L}[f] = \bar{\mathcal{L}}[f]$$

donde $\bar{\mathcal{L}}$ es el operador ADJUNTO a $\mathcal{L}$

$$\bar{\mathcal{L}}[f] = D^2(a_2^*[f]) - D(a_1^*[f]) + a_0^*[f] = \ddot{a}_2^* f + \dot{a}_2^* Df + \dot{a}_2^* Df + a_2^* D^2 f - \dot{a}_1^* - a_1^* Df + a_0^* f$$

$$= a_2^* D^2 f + (2\dot{a}_2^* - a_1^*) Df + (\ddot{a}_2^* - \dot{a}_1^* + a_0^*) f$$

si $\mathcal{L}(f) = \bar{\mathcal{L}}(f)$, los coeficientes de las derivadas deben ser iguales

1. $a_2(x) = a_2^*(x) \longrightarrow$ $\boxed{a_2(x) \in \mathbb{R}}$

2. $a_1(x) = 2\dot{a}_2^*(x) - a_1^*(x) \longrightarrow a_1(x) + a_1^*(x) = 2a_2(x)$

pero $a_1 + a_1^* \in \mathbb{R}$,

$$2a_1(x) = 2\dot{a}_2(x) \longrightarrow \boxed{a_1(x) = \dot{a}_2(x)}$$

3. $a_0(x) = \ddot{a}_2(x) + \dot{a}_1(x) + a_0(x) \rightarrow a_0(x) = a_0(x)$ trivial

Así, para $\mathcal{L}$ autoadjunto,

$$\mathcal{L}[\cdot] = a_2 D^2[\cdot] + \dot{a}_2 D[\cdot] + a_0[\cdot] = D\left(a_2(x) D[\cdot]\right) + a_0[\cdot]$$

Teorema

Siempre es posible transformar una ecuación diferencial lineal de 2do orden, con coeficientes reales, en autoadjunta

Sea $p(x) \in \mathbb{R}$

$$\mathcal{H} = p(x)\mathcal{L} = \underbrace{pa_2}_{\alpha_2} D^2 + \underbrace{pa_1}_{\alpha_1} D + \underbrace{pa_0}_{\alpha_0} = D(\alpha_2 D) + \alpha_0$$

así, $\dot{\alpha}_2 = \alpha_1$ y

$$(pa_2)' = pa_1$$
$$\dot{p}a_2 + p\dot{a}_2 = pa_1 \rightarrow \frac{dp}{dx}a_2 + p\frac{da_2}{dx} = pa_1 \cdot \frac{dx}{pa_2} \rightarrow \frac{dp}{p} + \frac{da_2}{a_2} = \frac{a_1}{a_2}dx$$

$$\int \frac{dp}{p} + \int \frac{da_2}{a_2} = \int \frac{a_1}{a_2}dx \rightarrow \ln(p) + \ln(a) = \int \frac{a_1}{a_2}dx \rightarrow p(x)\,a_2(x) = e^{\int \frac{a_1}{a_2}dx}$$

en general,

$$p(x) = \frac{1}{a_2(x)} e^{\int \frac{a_1}{a_2}dx}$$

multiplicar por $p(x)$ garantiza que un operador es autoadjunto

Operador Hermítico

Para un operador autoadjunto, se cumple

$$(\mathcal{L}f, g) = (f, \bar{\mathcal{L}}g) + \left\{ a_2^* W(f^*, g) \right\}_a^b$$

Un operador se dice **Hermítico** si

$$(\mathcal{L}f, g) = (f, \bar{\mathcal{L}}g)$$

es decir,

$$\left\{ a_2^* W(f^*, g) \right\}_a^b = 0$$

Ecuación de Sturm-Liouville

$$\mathcal{L}[f(x)] = h(x)$$

Tomando $\mathcal{H}[\,] = p(x)\mathcal{L}[\,] = D(p(x)a_2(x)D[\,]) + p(x)a_0(x)[\,]$ un operador hermítico,

$$\mathcal{H}[f(x)] = p(x)h(x) \rightarrow \mathcal{H}[f(x)] - p(x)h(x) = 0$$

Si tenemos una ecuación de autovalores $\mathcal{L}[f(x)] = -\lambda f(x)$,

$$\mathcal{H}[f(x)] + \lambda p(x) f(x) = 0$$

$$\hookrightarrow D(\underbrace{pa_2}_{q(x)} D[f]) + \underbrace{pa_0}_{r(x)}[f] + \lambda p f = 0$$

$$\rightarrow \frac{d}{dx}\left[q(x)\frac{d}{dx}f(x)\right] + r(x)f(x) + \lambda p(x) f(x) = 0$$

Ecuación de Sturm-Liouville

y $\frac{d^2}{dx^2}f(x) = -k^2 f(x)$

$q = 1$
$r = 0 \rightarrow f'' + k^2 f = 0 \rightarrow m^2 + k^2 = 0 \rightarrow m = \pm ik$
$p = 1$

$$f = A\sin(kx) + B\cos(kx)$$

con condiciones de cauchy, $k_n = \frac{n\pi}{L}$ → autovalores de la ecuación

$\rightarrow f_n(x) = A\sin\left(\frac{n\pi x}{L}\right)$ son las autofunciones

Ahora, podemos escribir

$$\mathcal{H}+\lambda p=0$$

① $\mathcal{H} f_m + \lambda_m p f_m = 0$

② $(\mathcal{H} f_n)^* - \lambda_n p f_n^* = 0$

$f_n^* \cdot ① - f_m \cdot ② \rightarrow f_n^* \mathcal{H} f_m - f_m (\mathcal{H} f_n)^* = (-\lambda_m + \lambda_n^*) p f_n^* f_m$

Integramos,

$$\int dx \left[f_n^* \mathcal{H} f_m - f_m (\mathcal{H} f_n)^*\right] = \int dx \left[(-\lambda_m + \lambda_n^*) p f_n^* f_m\right]$$

$$(f_n, \mathcal{H} f_m) - (\mathcal{H} f_n, f_m) = (-\lambda_m + \lambda_n^*) \int p f_n^* f_m \, dx$$

pero $(f_n, \mathcal{H} f_m) = (\mathcal{H} f_n, f_m)$ porque $\mathcal{H}$ es hermítico, entonces

$$(-\lambda_m + \lambda_n^*) \int p f_n^* f_m \, dx = 0$$

si $m=n$, $(\lambda_n^* - \lambda_n) \int p |f|^2 dx = 0$, pero $\int p |f|^2 dx \neq 0$

↳ $\lambda_n = \lambda_n^* \rightarrow \lambda_n \in \mathbb{R}$

$$(-\lambda_m + \lambda_n) \int p f_n^* f_m \, dx = 0 \longrightarrow \left(\sqrt{p(x)} f_n(x), \sqrt{p(x)} f_m(x)\right) = 0$$

En conclusión,

1. Los autovalores de una ecuación SL son reales
2. $\left\{\sqrt{p(x)} f_n(x)\right\}_{n=1}^{\infty}$ es una base ortonormal de funciones.

Ecuaciones Típicas en la Física

	$q(x)$	$r(x)$	λ	$p(x)$	(a,b)
• Legendre	$1-x^2$	0	$l(l+1)$	1	$(-1,1)$
• Legendre asociada	$1-x^2$	$-m^2/(1-x^2)$	$l(l+1)$	1	$(-1,1)$
• Chebyshev I	$(1-x^2)^{1/2}$	0	n^2	$1/(1-x^2)^{1/2}$	$(-1,1)$
• Chebyshev II	$(1-x^2)^{3/2}$	0	$n(n+2)$	$(1-x^2)^{1/2}$	$(-1,1)$
• Bessel	x	$-n^2/x$	k^2	x	$(0,b)$
• Laguerre	xe^{-x}	0	n	e^{-x}	$(0,\infty)$
• Laguerre asociada	$x^{k+1}e^{-x}$	0	n	$x^k e^{-x}$	$(0,\infty)$
• Hermite	e^{-x^2}	0	2α	e^{-x^2}	$(-\infty,\infty)$
• Oscilador	1	0	ω^2	1	$(-\infty,\infty)$

Funciones de Green

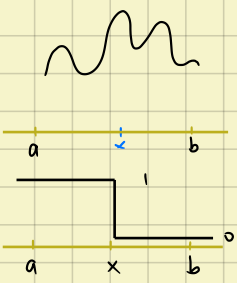
Consideremos

1. $\frac{dy}{dx} = i g(x) \rightarrow dy = i f dx \rightarrow \int_a^x dy = i \int_a^x f \, dx'$, $y(a) = y_0$

$$y(x) - y_0 = i \int_a^x f \, dx' \rightarrow y(x) = y_0 + i \int_a^x f \, dx'$$

$$= y_0 + i \int_a^b \theta(x-x') f(x') dx'$$

$$\rightarrow y(x) = y_0 + \int_a^b \underbrace{i \theta(x-x')}_{\text{FUNCIÓN DE GREEN}} f(x') dx'$$



Multiplicamos por la función de Heaviside para eliminar la parte que sobra de la solución

2 $\frac{d^2y}{dx^2} = f(x)$, $y(a) = y_0$, $\frac{dy(a)}{dx} = \bar{y}_0$

$\frac{d}{dx}\dot{y} = f(x)$

$\dot{y}(x) - \dot{y}(a) = \int_a^x f(x')\,dx'$

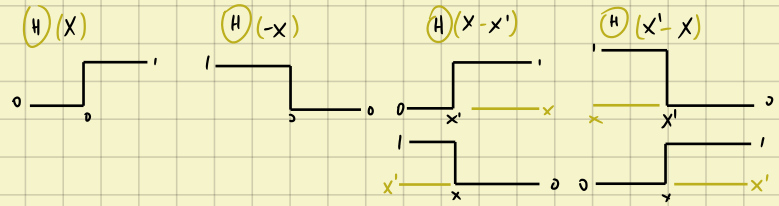
$y(x) - y(0) = \int_0^x \bar{y}_0\,dx' + \int_a^x \int_a^{x''} f(x')\,dx'\,dx''$

$y(x) = y_0 + \bar{y}_0(x-a) + \int_a^x \left(\int_0^{x''} dx' f(x') \right) dx''$

$= y_0 + \bar{y}_0(x-a) + \int_a^x f(x')(x-x')\,dx'$

$= y_0 + \bar{y}_0(x-a) + \int_a^b f(x')(x-x')\,\mathcal{U}(x-x')\,dx'$, tal que $G(x-x') = (x-x')\,\mathcal{U}(x-x')$

A partir de ahora, $\textcircled{H} = \mathcal{U}$ será la función de Heaviside o función escalón



Usamos la función $\textcircled{H}$ para poder cambiar el límite superior y truncar el exreso

$y(x) = y(a) + \int_a^x f(x')\,dx' \iff y(x) = y(a) + \int_a^b \textcircled{H}(x-x')\,f(x')\,dx'$

El KERNEL de la integral (en este caso $\textcircled{H}(x-x')$) es la FUNCIÓN DE GREEN asociada a la ecuación

En general,

$$y(x) = \alpha + \beta(x-a) + \int_a^b f(x')\,(x-x')\,\textcircled{H}(x-x')\,dx'$$

es la solución para un operador de 2do orden

ejemplo $x \in [0,1]$, $y(0) = y_0$ y $y(1) = y_1$

$\to y(0) = \alpha - \int_0^1 dx'\,(x')\,\textcircled{H}(-x')\,f(x') \to y(0) = \alpha$ (lo cancela $\textcircled{H}$)

$y(1) = y_0 + \beta + \int_0^1 dx'\,(1-x')\,\textcircled{H}(1-x')\,f(x') = y_0 + \beta + \int_0^1 dx'\,(1-x')\,f(x')$

$\to \beta = y_1 - y_0 - \int_0^1 (1-x')\,f(x')\,dx'$

tal que

$y(x) = y_0 + [(y_1 - y_0) - \int_0^1 (1-x')\,f(x')\,dx']\,x + \int_0^1 (x-x')\,\textcircled{H}(x-x')\,f(x')\,dx'$

$= y_0 + (y_1 - y_0)x + \int_0^1 \underbrace{\{-(1-x')x + (x-x')\,\textcircled{H}(x-x')\}}_{G(x-x')}\,f(x')\,dx'$

$$G(x-x') = \begin{cases} x'(x-1), & 0 \le x' \le x \\ x(x'-1), & x \le x' \le 1 \end{cases} = xx' - x'\,\textcircled{H}(x-x') - x\,\textcircled{H}(x'-x)$$

Para ecuaciones SL

$\mathcal{L}[y(x)] = f(x) \to \mathcal{L}[\phi_n] = \lambda_n \phi_n$

Propongamos $y(x) = \sum_n^\infty \alpha_n \phi_n(x)$; $f(x) = \sum_n^\infty \beta_n \phi_n(x)$

$\to \mathcal{L}\left[\sum_n \alpha_n \phi_n\right] = \sum_n \beta_n \phi_n$

$\sum_n \alpha_n \mathcal{L}[\phi_n] = \sum_n \beta_n \phi_n$

$\sum_n \alpha_n \lambda_n \phi_n = \sum_n \beta_n \phi_n \to \sum_n \underbrace{(\alpha_n \lambda_n - \beta_n)}_{0, \text{ por L.I}} \phi_n(x) = 0$

$\therefore \alpha_n \lambda_n - \beta_n = 0 \to \alpha_n = \frac{\beta_n}{\lambda_n}$

pero $\beta_n = \int_a^b \phi_n^*(x) f(x)\, dx$

$\to y(x) = \sum_{n=1}^\infty \left(\int_a^b \phi_n^*(x') f(x') dx'\right) \cdot \frac{1}{\lambda_n} \phi_n(x) = \int_a^b f(x') \underbrace{\left[\sum_n \frac{\phi_n^*(x')\phi_n(x)}{\lambda_n}\right]}_{G(x-x')} dx'$

ejemplo $\mathcal{D}[\cdot] = f(x)$, $x \in [0,1]$, $y(0) = y(1) = 0$

Sea $\mathcal{L}[\phi_n(x)] = -\lambda_n \phi_n(x)$

$\frac{d^2}{dx^2}\phi_n(x) = -\lambda \phi_n(x) \to \left(\frac{d^2}{dx^2} + \lambda\right)\phi_n(x) = 0$ $\left[m^2 + \lambda = 0 \to m = \pm\sqrt{\lambda}\, i\right]$

$\to \phi_n(x) = A\sin(mx) + B\cos(mx)$

$\phi_n(0) = B = 0 \to \phi_n(x) = A\sin(mx)$

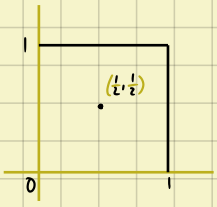
$\phi_n(1) = A\sin(m) = 0 \to \sin(m) = 0 \to m = n\pi \to -\lambda = n^2\pi^2$

$\lambda = -n^2\pi^2$

normalizamos,

$A^2 \int_0^1 \phi_n(x)\, \phi_n(x)\, dx = 1$

$\to A = \sqrt{2}$



Construimos la función de Green

$G(x,x') = \sum_n \frac{\sqrt{2}\sin(n\pi x)\sqrt{2}\sin(n\pi x')}{-n^2\pi^2} = -\frac{2}{\pi^2}\sum_n \frac{\sin(n\pi x)\sin(n\pi x')}{n^2}$

De antes,

$G(x,x') = xx' - x'\,H(x-x') - x\,H(x'-x)$

Evaluemos en el punto central

$G(\frac{1}{2},\frac{1}{2}) = \frac{1}{4} - \frac{1}{2}H(0) - \frac{1}{2}H(0) = -\frac{1}{4}$

$G(\frac{1}{2},\frac{1}{2}) = -\frac{2}{\pi^2}\sum \frac{\sin^2(n\frac{\pi}{2})}{n^2} = -\frac{2}{\pi^2}\sum_n \frac{1}{n^2} = -\frac{2}{\pi^2}\frac{\pi^2}{8} = -\frac{1}{4}$

lo cual muestra que ambas representaciones son iguales.

FUNCIÓN SIGNO

$\operatorname{sgn}(x) = \begin{cases} +1, & x > 0 \\ -1, & x < 0 \end{cases}$

pero $\operatorname{sgn}(x) = H(x) - H(-x)$

$\to \frac{d}{dx}\operatorname{sgn}(x) = \frac{d}{dx}H(x) - \frac{d}{dx}H(-x) = \delta(x) + \delta(-x) = 2\delta(x)$

$|x| = \begin{cases} x, & x > 0 \\ -x, & x < 0 \end{cases}$

$\star$: $\frac{d}{dx}H(x) = \frac{d}{dx}\eta(x) = \delta(x)$, dirac

tal que $|x| = x\,\mathrm{sgn}(x)$. Veamos

$$\frac{d^2}{dx^2}|x| = \frac{d}{dx}\left[\frac{d}{dx}(x\,\mathrm{sgn}(x))\right] = \frac{d}{dx}\left[\mathrm{sgn}(x) + x\,2\delta(x)\right] = 2\delta(x) + \frac{d}{dx}\underbrace{(2x\,\delta(x))}_{0} = 2\delta(x)$$

Considerando

$$\frac{d^2y(x)}{dx^2} = f(x) \rightarrow y(x) = \int G(x,x')f(x')\,dx'$$

para $f(x') = \delta(x-x'')$

$$y(x) = G(x-x'') \rightarrow \frac{d^2}{dx^2}G(x,x') = \delta(x-x'), \text{ además, } \delta(y-y') = \frac{d}{dx}\left(\frac{1}{2}(x-x')\right) \rightarrow \text{de lo anterior}$$

$$\rightarrow G(x-x') = \frac{1}{2}|x-x'|$$

tal que

$$y(x) = \int \frac{1}{2}|x-x'|f(x')\,dx'$$

En general, para $\mathcal{L}[y] = f$, si $f = \delta$, entonces $y = G$. Esto nos permite hallar expresiones equivalentes para G.

3. Funciones Especiales

Funciones de Bessel

Consideremos $\nabla^2\psi(\vec{r}) = 0$ en coordenadas cilíndricas. Asumiendo una separación $\psi(\vec{r}) = R(r)\,\Phi(\phi)\,Z(z)$

$$\begin{cases} \ddot{\Phi} = -m^2\Phi \\ \ddot{Z} = k^2 Z \\ \frac{1}{\rho}\frac{d}{d\rho}\left(\rho\frac{dR}{d\rho}\right) + \left(k^2 - \frac{m^2}{\rho^2}\right)R = 0 \end{cases}$$

sustituyendo $x = k\rho$ en la ecuación 3,

$$\frac{k^2}{x}\frac{d}{dx}\left(x\frac{d}{dx}R\left(\frac{x}{k}\right)\right) + \left(1 - \frac{m^2}{x^2}\right)R = 0$$

$$\frac{1}{x}\left(\dot{R} + x\ddot{R}\right) + \left(1 - \frac{m^2}{x^2}\right)R = 0$$

$$x^2\ddot{R} + x\dot{R} + (x^2 - m^2)R = 0$$

proponemos $y(x) = \sum_{n=0}^{\infty} a_n x^{n+s}$, para $y = R$. Agregamos s para tener una potencia mínima

$$\rightarrow \dot{y} = \sum (n+s)\,a_n x^{n+s-1}$$

$$\ddot{y} = \sum (n+s-1)(n+s)\,a_n x^{n+s-1}$$

$$\therefore \sum_0^{\infty}\left[(n+s)(n+s-1)a_n + (n+s)a_n - m^2 a_n\right]x^{n+s} + \sum_0^{\infty} a_n x^{n+s+2} = 0$$

$$\sum_0^{\infty}\left\{(n+s)\left[(n+s-1)+1\right] - m^2\right\}a_n x^{n+s} + \sum_0^{\infty} a_n x^{n+s+2} = 0$$

$$\sum_0^{\infty}\left[(n+s)^2 - m^2\right]a_n x^{n+s} + \sum_0^{\infty} a_n x^{n+s} = 0, \text{ sustituimos } n = n'+2$$

$$\sum_{-2}^{\infty}\left[(n'+2+s)^2 - m^2\right]a_{n'+2}x^{n'+2+s} + \sum_0^{\infty} a_n x^{n+s} = 0$$

$$\underset{n=0}{(s^2 - m^2)a_0 x^s} + \underset{n=1}{\left[(1+s)^2 - m^2\right]a_1 x^{s+1}} + \sum_0^{\infty}\left\{\left[(n+2+s)^2 - m^2\right]a_{n+2} + a_n\right\}x^{n+s} = 0$$

Ya que $\{x^n\}_0^{\infty}$ es LI,

1. $(s^2 - m^2)a_0 = 0$, pero $a_0 \neq 0 \rightarrow s = \pm m$

$2\left[(1+s)^2 - m^2\right]a_1 = 0 \rightarrow (1+2s+s^2-m^2)a_1 = 0$

$1+2s=0 \rightarrow s = m = -\frac{1}{2}$ ó $a_1 = 0$

Por ende, la serie es par $(a_0, a_2, a_4, \dots)$ con $a_{n+2} = \frac{1}{(n+2)(n+2m+2)} a_n$, tal que

$n=0 \quad a_2 = -\frac{1}{2(2(m+1))} a_0$

$a_4 = -\frac{a_2}{4(2m+4)} = \frac{a_0}{2 \cdot 2^4 (m+2)(m+1)}$

$$a_{2k} = \frac{(-1)^k a_0 \Gamma(m+1)}{2^{2k} \Gamma(m+k+1)}$$

* Recordemos $\Gamma(\nu) = \int x^{\nu-1} e^{-x} dx$

- $\Gamma(\nu+1) = \nu\Gamma(\nu)$
- Si $\nu = n \in \mathbb{N} \cup \{0\}$, $\Gamma(n+1) = n\Gamma(n) = n!$

$\rightarrow a_{2k} = \frac{(-1)^k a_0 m!}{2^{2k} k!(m+k)!}$, tal que

$$y(x) = \sum_{k=0}^{\infty} \frac{(-1)^k a_0 m!}{2^{2k} k!(m+k)!} \cdot x^{2k+m}$$

$$= a_0 \Gamma(m+1) 2^m \sum \frac{(-1)^k}{k! \Gamma(k+m+1)} \left(\frac{x}{2}\right)^{m+2k}$$

Así, definimos la FUNCIÓN DE BESSEL de primera clase como

$$J_\nu(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(\nu+k+1)} \left(\frac{x}{2}\right)^{\nu+2k}$$

Si consideramos $s = -m$

$$a_{2k} = \frac{(-1)^k a_0 \Gamma(-m+1)}{2^{2k} k! \Gamma(k-m+1)}$$

$$\rightarrow J_{-\nu}(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(-\nu+k+1)} \left(\frac{x}{2}\right)^{-\nu+2k}$$

de forma entera,

$$J_{-n} = \sum_{k=n}^{\infty} \frac{(-1)^k}{k!(k-n)!} \left(\frac{x}{2}\right)^{2k-n} = \sum_{k'=0}^{\infty} \frac{(-1)^{k'+n}}{(k'+n)! k'!} \left(\frac{x}{2}\right)^{2k'+n} \quad \text{con } k' = k-n$$

$$= (-1)^n \underbrace{\sum_{k'=0}^{\infty} \frac{(-1)^{k'}}{(k'+n)! k'!} \left(\frac{x}{2}\right)^{n+2k'}}_{J_n(x)}$$

$$\rightarrow J_{-n}(x) = (-1)^n J_n(x)$$

No obstante,

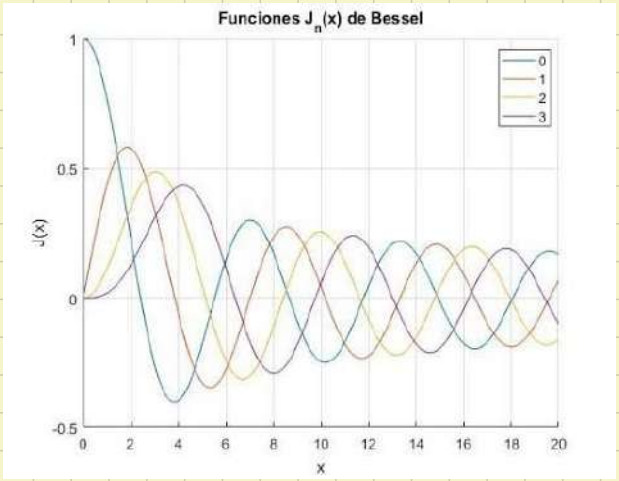
$$y(x) = A J_\nu(x) + B N_\nu(x)$$

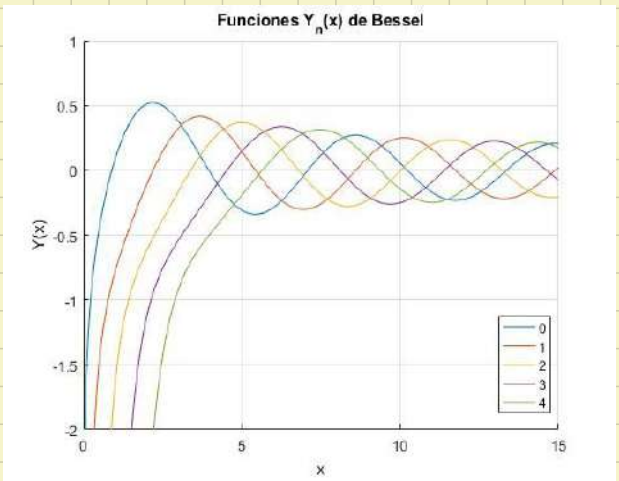
Definimos las FUNCIONES DE NEUMANN

$$N_\nu(x) = \frac{1}{\pi}\left[\frac{\partial J_\nu(x)}{\partial \nu} - (-1)^\nu \frac{\partial J_{-\nu}(x)}{\partial \nu}\right]_{\nu=n}$$

Tanto $J_\nu(x)$ como $N_\nu(x)$ satisfacen

$$\frac{d^2}{dx^2}[\cdot] + \frac{1}{x}\frac{d}{dx}[\cdot] + \left(1 - \frac{n^2}{x^2}\right)[\cdot] = 0$$





Funciones Generadoras

Una función generadora es aquella que permite expandir una serie de polinomios asociados. Veamos,

$$g(x,t) = e^{\frac{x}{2}\left(t-\frac{1}{t}\right)}$$

$$= e^{\frac{x}{2}t}\, e^{-\frac{x}{2t}} = \left(\sum_{r=0}^{\infty}\frac{\left(\frac{x}{2}t\right)^r}{r!}\right)\left(\sum_{s=0}^{\infty}\frac{\left(-\frac{x}{2t}\right)^s}{s!}\right) = \left(\sum_{r=0}^{\infty}\left(\frac{x}{2}\right)^r\frac{t^r}{r!}\right)\left(\sum_{s=0}^{\infty}(-1)^s\left(\frac{x}{2}\right)^s\frac{t^{-s}}{s!}\right), \text{ sea } n = r-s$$

$$= \left(\sum_{n=0}^{\infty}\left(\frac{x}{2}\right)^{n-s}\frac{t^{n-s}}{(n-s)!}\right)\left(\sum_{s=0}^{\infty}(-1)^s\left(\frac{x}{2}\right)^s\frac{t^{-s}}{s!}\right) = \sum_n\left[\sum_s\left(\frac{x}{2}\right)^{n-2s}\frac{(-1)^s}{(n-s)!\,s!}\right]t^n$$

$$= \sum_{n=0}^{\infty} J_n(x)\, t^n \quad \rightarrow \text{sirve para generar } J_n$$

(J_n)

- **Relación de recurrencia**

1. $\frac{\partial g}{\partial t} = g\,\frac{x}{2}\left(1+\frac{1}{t^2}\right) = \sum_n J_n\, n t^{n-1}$

$$\rightarrow \sum_n J_n t^n\left(1+\frac{1}{t^2}\right) = \frac{2}{x}\sum_n J_n\, n t^{n-1}$$

$$\sum_n\left(J_n t^n + J_n t^{n-2}\right) = \sum_n \frac{2n}{x} J_n t^{n-1}$$

$$\rightarrow J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x}J_n(x) \quad \text{ó} \quad \boxed{J_{n+1}(x) = \frac{2n}{x}J_n(x) - J_{n-1}(x)}$$

2. $\frac{\partial g}{\partial x} = g\,\frac{1}{2}\left(t-\frac{1}{t}\right) = \sum_n J_n'(x)\, t^n$

$$\rightarrow \sum_n J_n t^n \frac{1}{2}\left(t-\frac{1}{t}\right) = \sum_n J_n' t^n$$

$$\sum_n\left(J_n t^{n+1} - J_n t^{n-1}\right) = \sum_n 2J_n' t^n$$

$$\rightarrow \boxed{J_{n-1}(x) - J_{n+1}(x) = 2J_n'(x)}$$

$$\rightarrow \begin{cases} J_{n-1}(x) = \frac{n}{x}J_n(x) + \frac{dJ_n(x)}{dx} \\ J_{n+1}(x) = \frac{n}{x}J_n(x) - \frac{dJ_n(x)}{dx} \end{cases} \rightarrow \begin{cases} J_{n+1}(x) = \tilde{N}_n^+ J_n(x) \\ J_{n-1}(x) = \tilde{N}_n^- J_n(x) \end{cases}, \quad \tilde{N}_n^{\pm}[\cdot] = \left(\frac{n}{x} \mp \frac{d}{dx}\right)[\cdot]$$

Consideremos $t = e^{i\theta}$

$$g(x,t) = e^{\frac{x}{2}\left(t-\frac{1}{t}\right)} = e^{\frac{x}{2}\left(e^{i\theta}-e^{-i\theta}\right)} = e^{\frac{x}{2}2i\left(\frac{e^{i\theta}-e^{-i\theta}}{2i}\right)} = e^{ix\sin(\theta)} = \sum_{-\infty}^{\infty} J_n t^n = \sum_{-\infty}^{\infty} J_n e^{in\theta} = \sum_0^{\infty}\left(J_n e^{in\theta} + J_{-n}e^{-in\theta}\right) = \sum_0^{\infty}\left[J_n e^{in\theta} + (-1)^n J_n e^{-in\theta}\right]$$

$$= \sum_0^{\infty} J_n(x)\left(e^{in\theta} + (-1)^n e^{-in\theta}\right) = J_0 + 2iJ_1\left(\frac{e^{i\theta}-e^{-i\theta}}{2i}\right) + 2J_2\left(\frac{e^{i2\theta}+e^{-i2\theta}}{2}\right) +$$

$$\rightarrow \cos(x\sin\theta) + i\sin(x\sin\theta) = J_0(x) + 2\sum_{n=1}^{\infty} J_{2n}(x)\cos(2n\theta) + 2i\sum_{n=1}^{\infty} J_{2n-1}(x)\sin((2n-1)\theta)$$

proyectando sobre la base completa $\{\cos n\theta, \sin n\theta\}_{n=1}^{\infty}$

$$J_n(x) = \frac{1}{\pi}\int_0^{\pi}\left[\cos(n\theta)\cos(x\sin\theta) + \sin(n\theta)\sin(x\sin\theta)\right]d\theta$$

Podemos tener recurrencia para $N_n(x)$

1. $N_{n-1}(x) + N_{n+1}(x) = \frac{2n}{x}N_n(x)$
2. $N_{n-1}(x) - N_{n+1}(x) = 2N_n'(x)$

1+2 $N_{n-1}(x) = \left(\frac{n}{x}+\frac{d}{dx}\right)N_n(x) = \tilde{N}_-[N_n(x)]$

1-2 $N_{n+1}(x) = \left(\frac{n}{x}-\frac{d}{dx}\right)N_n(x) = \tilde{N}_+[N_n(x)]$

y son las mismas que para $J_n(x)$

Ortogonalidad

Sustituyendo $x = k\rho$ en la ED de Bessel,

$$\rho^2\frac{d^2}{d\rho^2}R(k\rho) + \rho\frac{d}{d\rho}R(k\rho) + (k^2\rho^2 - n^2)R(k\rho) = 0$$

Pasamos a forma autoadjunta,

$$\frac{d}{d\rho}\left[\rho \frac{dR(k\rho)}{d\rho}\right] + (k^2\rho^2 - n^2) R(k\rho) = 0$$

Así, tenemos que $J_\nu\left(\alpha_{\nu m}\frac{\rho}{a}\right)$ es una base ortogonal en $\rho \in [0, a]$, para $\alpha_{\nu m}$ los ceros de J_ν.

$$\int_0^a J_\nu\left(\alpha_{\nu m}\frac{\rho}{a}\right) J_\nu(\alpha_{\nu n})\, \rho\, d\rho = 0$$

Para normalizar,

$$A^2 \int_0^a \left[J_\nu\left(\alpha_{\nu m}\frac{\rho}{a}\right)\right]^2 \rho\, d\rho = 1$$

$$\rightarrow A = \frac{\sqrt{2}}{a}\frac{1}{J_{\nu+1}(\alpha_{\nu m})} \quad \rightarrow J_{\nu+1} \text{ evaluada en el cero de } J_\nu$$

tal que

$$\left\{ J_\nu^N\left(\alpha_{\nu m}\frac{\rho}{a}\right) = \frac{\sqrt{2}}{a}\frac{1}{J_{\nu+1}(\alpha_{\nu m})} J_\nu\left(\alpha_{\nu m}\frac{\rho}{a}\right) \right\}$$

es una base ortonormal

Para una base continua,

$$\int_0^\infty \rho J_\nu(k\rho) J_\nu(k'\rho)\, d\rho = \frac{1}{k}\delta(k-k')$$

$$\int_{-\infty}^\infty k J_\nu(k\rho) J_\nu(k\rho')\, dk = \frac{1}{\rho}\delta(\rho-\rho')$$

Funciones de Hankel

$$H_\nu^{(1)}(x) = J_\nu(x) + i N_\nu(x)$$

$$H_\nu^{(2)}(x) = J_\nu(x) - i N_\nu(x)$$

Cumplen las mismas propiedades que $J_\nu(x)$

Funciones Esféricas de Bessel

Para solucionar la eq de Helmholtz, sustituimos $R(\rho) = \frac{Z(\rho)}{\rho^{1/2}}$

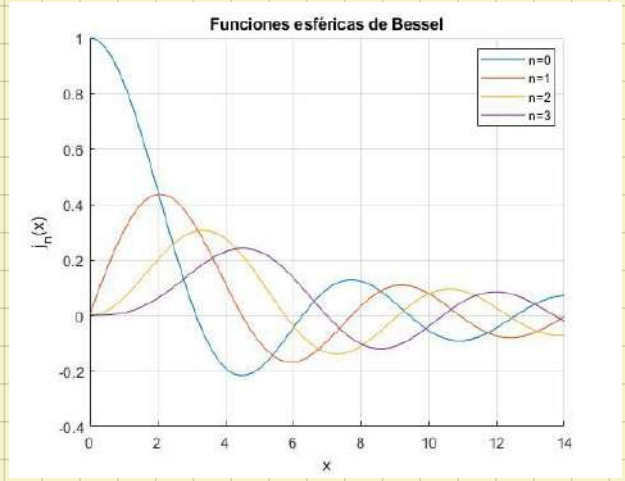
$$\frac{d}{dr}\left[r\frac{dZ}{dr}\right] + \left[k^2 r - \frac{(n+\frac{1}{2})^2}{r}\right] Z = 0$$

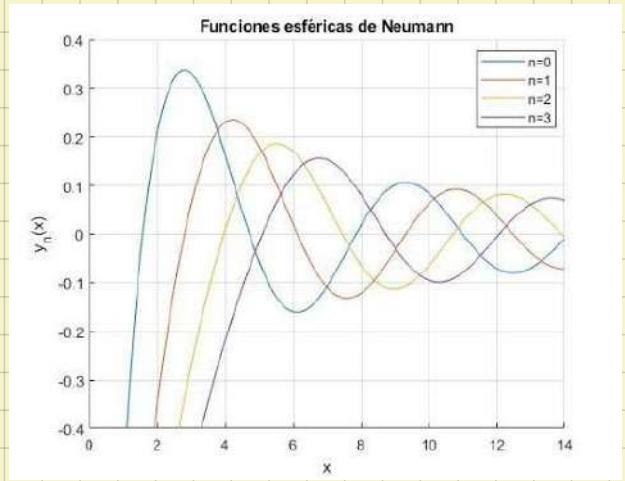
de donde

$$j_n(x) = \sqrt{\frac{\pi}{2x}} J_{n+\frac{1}{2}}(x)$$

$$n_n(x) = \sqrt{\frac{\pi}{2x}} N_{n+\frac{1}{2}}(x)$$

$$\begin{cases} h_n^{(1)}(x) = j_n(x) + i n_n(x) \\ h_n^{(2)}(x) = j_n(x) - i n_n(x) \end{cases}$$





Partícula libre en mecánica cuántica

$$\hat{H}\Psi(\vec{r}) = E\Psi(\vec{r})$$

$$(T+V)\Psi(\vec{r}) = E\Psi(\vec{r})$$

$$\left[-\frac{\hbar^2}{2m}\vec{\nabla}^2 + V(\vec{r}) - E\right]\Psi(\vec{r}) = 0 \rightarrow \left[\vec{\nabla}^2 - \frac{2mV(\vec{r})}{\hbar^2} + \frac{2mE}{\hbar^2}\right]\Psi(\vec{r}) = 0$$

pero $T = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m} = E$ y $V(\vec{r}) = 0$ por ser libre

ahora, con

$$\begin{cases} p = \hbar k & \text{Broglie} \\ E = \hbar\omega & \text{Einstein} \end{cases} \rightarrow p = \frac{h}{\lambda} \rightarrow \lambda = \frac{h}{p}$$

$\rightarrow [\vec{\nabla}^2 + k^2]\,\psi(\vec{r}) = 0$, que es la ecuación de Helmholtz. Separamos las variables,

$$\begin{cases} \left(\frac{d^2}{dx^2} + k_x^2\right) X(x) = 0 \\ \left(\frac{d^2}{dy^2} + k_y^2\right) Y(y) = 0 \\ \left(\frac{d^2}{dz^2} + k_z^2\right) Z(z) = 0 \end{cases} \rightarrow \psi(\vec{r}) = e^{i(k_x x + k_y y + k_z z)} = e^{i(\vec{k}\cdot\vec{r})} \rightarrow \text{Onda Plana}$$

normalizando,

$$\psi^N(\vec{r}) = \frac{1}{(2\pi)^{3/2}} e^{i(\vec{k}\cdot\vec{r})}$$

En coordenadas esféricas,

$$L^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right]$$

$$-\vec{\nabla}^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) - \frac{\hat{L}^2}{r^2\hbar^2}$$

tal que

$$\psi(\vec{r}) = R(r)\,\underbrace{Y_\ell^m(\theta,\varphi)}_{\text{armónico esférico}}$$

$$\rightarrow \hat{L}^2 Y_\ell^m(\theta,\varphi) = \hbar^2 l(l+1) Y_\ell^m(\theta,\varphi)$$

con $R(r) = \frac{u(r)}{r}$

$$\frac{\partial^2}{\partial r^2}u(r) - \frac{l(l+1)}{r^2}u(r) - \left(\cancel{\frac{2m}{\hbar^2}V(r)}^{\text{partícula libre}} + \frac{2m}{\hbar^2}E\right)u(r) = 0$$

$$\frac{\partial^2}{\partial r^2}u - \frac{l(l+1)}{r^2}u + k^2 u = 0 \rightarrow l \text{ es un autovalor asociado al operador } \hat{L}^2,\ l \in \mathbb{N}$$

sustituimos $\rho = kr$, $\frac{d}{dr} = k\frac{d}{d\rho}$ y $\frac{d^2}{dr^2} = k^2\frac{d^2}{d\rho^2}$

$$\left[\cancel{k^2}\frac{\partial^2}{\partial\rho^2} - \cancel{k^2}\frac{l(l+1)}{\rho^2} + \cancel{k^2}\right] f_l(\rho)\cdot\rho = 0$$

$\rightarrow \left[\frac{d^2}{d\rho^2} + \frac{2}{\rho}\frac{d}{d\rho} + 1 - \frac{l(l+1)}{\rho^2}\right] f_l(\rho) = 0$, que es la ED esférica de Bessel

Con condiciones de contorno, la solución es

$$f_l(kr) = A_l\, j_l(kr)$$

teniendo en cuenta $j_0(x) = \frac{\sin x}{x}$, si $f(ka) = 0$

$$\frac{\sin(ka)}{ka} = 0 \rightarrow ka = n\pi \rightarrow k = \frac{n\pi}{a}$$

recordemos $E = \frac{\hbar^2 k^2}{2m}$. Por lo tanto, $E_n = \frac{\hbar^2 n^2\pi^2}{2ma^2}$, es decir, la energía en cuántica está discretizada.

Las soluciones se relacionan de la forma

$$e^{i(\vec{k}\cdot\vec{r})} = \sum_{l=0}^{\infty}(2l+1)\,i^l\, j_l(kr)\,\underbrace{P_l(\cos\theta)}_{\text{polinomios de Legendre}}$$

Polinomios de Legendre

Partiendo de la eq. de Laplace $\nabla^2 \psi(r,\theta,\varphi) = 0$ y separando como $\psi(r,\theta,\varphi) = \frac{u(r)}{r} P(\theta) T(\varphi)$

$$\frac{\sin\theta}{P(\theta)} \frac{\partial}{\partial\theta}\left(\sin\theta \frac{dP(\theta)}{\partial\theta}\right) + b\sin^2\theta = m^2$$

$$\frac{\partial}{\partial\theta}\left(\sin\theta \frac{\partial P(\theta)}{\partial\theta}\right) + b\sin\theta P(\theta) - \frac{m^2}{\sin\theta} P(\theta) = 0$$

sustituimos $x = \cos\theta$, $\sin\theta = \sqrt{1-x^2}$

$$-\sqrt{1-x^2}\frac{\partial}{\partial x}\left(-\sqrt{1-x^2}\sqrt{1-x^2}\frac{\partial}{\partial x}P(x)\right) + b\sqrt{1-x^2}P(x) - \frac{m^2}{\sqrt{1-x^2}}P(x) = 0$$

$$\frac{\partial}{\partial x}\left((1-x^2)\frac{\partial}{\partial x}P(x)\right) + bP(x) - \frac{m^2}{1-x^2}P(x) = 0$$

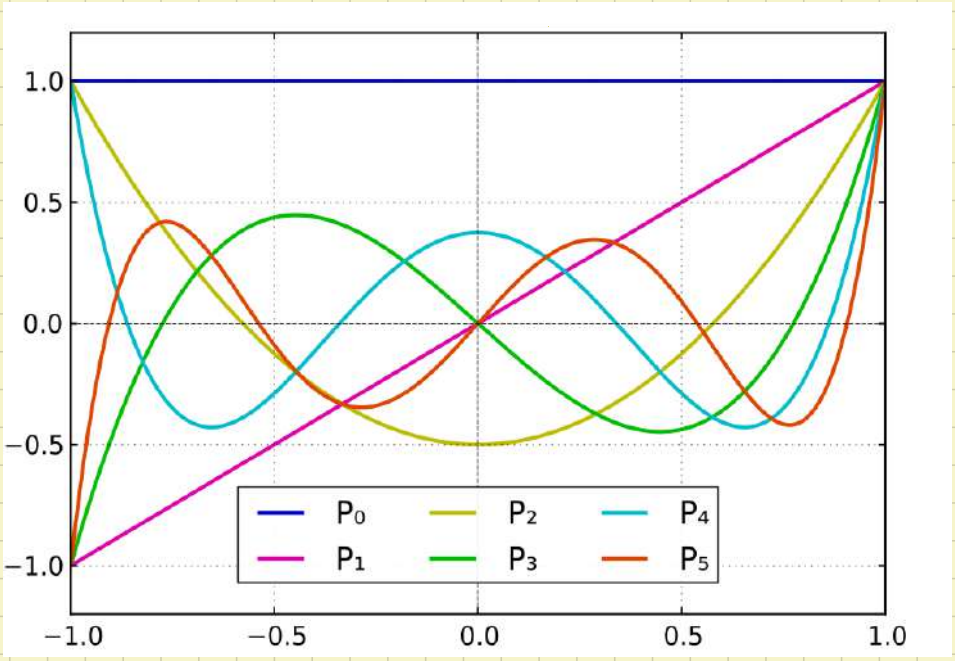
Si $m=0$, obtenemos la EDO asociada a los polinomios de Legendre

$$\frac{d}{dx}\left[(1-x^2)\frac{d}{dx}P(x)\right] + bP(x) = 0 \quad \text{ó} \quad (1-x^2)\frac{d^2}{dx^2}P(x) - 2x\frac{d}{dx}P(x) + bP(x) = 0, \text{ con } b = l(l+1)$$

Aplicando el método de Frobenius,

$$P_l(x) = \sum_{k=0}^{\lfloor l/2 \rfloor} \frac{(-1)^k (2l-2k)!}{2^l k!(l-k)!(l-2k)!} x^{l-2k}$$

$$\begin{aligned}
P_0(x) &= 1;\\
P_1(x) &= x;\\
P_2(x) &= \frac{1}{2}(3x^2-1);\\
P_3(x) &= \frac{1}{2}(5x^3-3x);\\
P_4(x) &= \frac{1}{8}(35x^4-30x^2+3);\\
P_5(x) &= \frac{1}{8}(63x^5-70x^3+15x);\\
P_6(x) &= \frac{1}{16}(231x^6-315x^4+105x^2-5).
\end{aligned}$$



También se pueden obtener aplicando Gram-Schmidt al conjunto de monomios $\{x^n\}_0^\infty$, en $x \in [-1,1]$

• Función Generadora

Consideremos un potencial Coulombiano

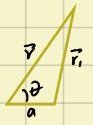
$$V(r) = \frac{1}{4\pi\varepsilon_0}\frac{q}{r_1} = \frac{q}{4\pi\varepsilon_0}\frac{1}{\sqrt{a^2+r^2-2ar\cos\theta}}$$

llamamos $\frac{a}{r} = t$ y $x = \cos\theta$

$$= \frac{q}{4\pi\varepsilon_0}\frac{1}{r}\frac{1}{\sqrt{1+t^2-2xt}}$$

$$= \frac{q}{4\pi\varepsilon_0}\frac{1}{r}\left(1-2(x+t)^{1/2}\right), \text{ expandiendo el monomio.}$$

$$= \frac{q}{4\pi\varepsilon_0}\frac{1}{r}\sum P_l(\cos\theta)\left(\frac{a}{r}\right)^l$$



ley del coseno
$r_1^2 = a^2 + r^2 - 2ar\cos\theta$

De forma que

$$G(t,x) = \frac{1}{\sqrt{1-2xt+t^2}} = \sum_{l=0}^{\infty} P_l(x)\, t^l$$

• Relaciones de Recurrencia

1. $(2n+1)xP_n = (n+1)P_{n+1} + nP_{n-1} \rightarrow P_{n+1} = \frac{2n+1}{n+1}xP_n - \frac{n}{n+1}P_{n-1}$
2. $P_n' + P_{n-2}' = 2xP_{n-1}' + P_{n-1}$
3. $P_{n+1}' - P_{n-1}' = (2n+1)P_n$
4. $P_{n+1}' = (n+1)P_n + xP_n'$
5. $(1-x^2)P_n' = nP_{n-1} - nxP_n$
6. $(1-x^2)P_n' = (n+1)xP_n - (n+1)P_{n+1}$

- **Propiedades**

1. $P_l(-x) = (-1)^l P_l(x)$ → P_l tiene que ser par o impar
2. $\forall l,\ P_{2l+1}(0) = 0$
3. $\forall l,\ P_{2l}(0) = \frac{(-1)^l (2l)!}{2^{2l}(l!)^2}$

- **Ortogonalidad**

$$\int_{-1}^{1} [P_l(x)]^2 dx = \frac{2}{2l+1}$$

Para tener una base normalizada,

$$P_l^N(x) = \sqrt{\frac{2l+1}{2}} P_l(x)$$

4. Armónicos Esféricos

Polinomios Asociados de Legendre

Si partimos de la ED de Legendre para $m \neq 0$

$$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta \frac{\partial}{\partial\theta}[\cdot]\right) + \left[l(l+1) - \frac{m^2}{\sin^2\theta}\right][\cdot] = 0$$

haciendo $x = \cos\theta$

$$-\sqrt{1-x^2}\frac{\partial}{\partial x}\left(-\sqrt{1-x^2}\sqrt{1-x^2}\frac{\partial}{\partial x}P_l^m(x)\right) + b\sqrt{1-x^2}P_l^m(x) - \frac{m^2}{\sqrt{1-x^2}}P_l^m(x) = 0$$

$$\frac{\partial}{\partial x}\left((1-x^2)\frac{\partial}{\partial x}P_l^m(x)\right) + b P_l^m(x) - \frac{m^2}{1-x^2}P_l^m(x) = 0$$

$$(1-x^2)\frac{d^2 P_l^m(x)}{dx^2} - 2x\frac{dP_l^m(x)}{dx} + \left[l(l+1) - \frac{m^2}{1-x^2}\right]P_l^m(x) = 0$$

Ahora, tenemos las siguientes características:

- $P_l^m(x) = (1-x^2)^{m/2}\frac{d^m P_l(x)}{dx^m}$, con $-l \leq m \leq l$
- $P_l^0(x) = P_l(x)$
- $P_l(x) = \frac{1}{2^l l!}\frac{d^l}{dx^l}(x^2-1)^l$ → Fórmula de Rodrigues
- $P_l^m(x) = (1-x^2)^{2/m}\frac{d^m}{dx^m}\left[\frac{1}{2^l l!}\frac{d^l}{dx^l}(x^2-1)^l\right]$
- $P_l^{-m}(x) = (-1)^m\frac{(l-m)!}{(l+m)!}P_l^m(x)$ → también es solución de la ED

Para un l dado, tenemos $2l+1$ $P_l^m(x)$ asociados. La paridad de estos polinomios se invierte; para m fijo,

P_{2k}^m → es impar
P_{2k+1}^m → es par

$$\Big\} \quad P_l^m(-x) = (-1)^{l+m}P_l^m(x)$$

- **Función Generadora**

Con m fijo.

$$G(x,t) = \frac{(2m)!}{(2m)!!}\frac{(1-x^2)^{m/2}}{(1-2tx+t^2)^{m+1/2}} = \sum_{s=0}^{\infty} P_{s+m}^m(x)\,t^s$$

- **Relaciones de Recurrencia**

1. $(2l+1)\sqrt{1-x^2}\,P_l^m(x) = P_{l+1}^{m+1}(x) - P_{l-1}^{m+1}(x)$
2. $(2l+1)x P_l^m(x) = (l+m)P_{l-1}^m(x) + (l-m+1)P_{l+1}^m(x)$
3. $P_l^{m+1}(x)\ \frac{2mx}{(1-x^2)^{1/2}}P_l^m(x) + [l(l+1) - m(m-1)]P_l^{m-1} = 0$

Polinomios Asociados de Legendre

$$P_0^0(x) = 1$$

$$P_1^{-1}(x) = -\frac{1}{2}P_1^1(x)$$
$$P_1^0(x) = x$$
$$P_1^1(x) = (1-x^2)^{1/2}$$

$$P_2^{-2}(x) = \frac{1}{24}P_2^2(x)$$
$$P_2^{-1}(x) = -\frac{1}{6}P_2^1(x)$$
$$P_2^0(x) = \frac{1}{2}(3x^2-1)$$
$$P_2^1(x) = -3x(1-x^2)^{1/2}$$
$$P_2^2(x) = 3(1-x^2)$$

$$P_3^{-3}(x) = -\frac{1}{720}P_3^3(x)$$
$$P_3^{-2}(x) = \frac{1}{120}P_3^2(x)$$
$$P_3^{-1}(x) = -\frac{1}{12}P_3^1(x)$$
$$P_3^0(x) = \frac{1}{2}(5x^3-3x)$$
$$P_3^1(x) = -\frac{3}{2}(5x^2-1)(1-x^2)^{1/2}$$
$$P_3^2(x) = 15x(1-x^2)$$
$$P_3^3(x) = -15(1-x^2)^{3/2}$$

$$P_4^{-4}(x) = \frac{1}{40320}P_4^4(x)$$
$$P_4^{-3}(x) = -\frac{1}{5040}P_4^3(x)$$
$$P_4^{-2}(x) = \frac{1}{360}P_4^2(x)$$
$$P_4^{-1}(x) = -\frac{1}{20}P_4^1(x)$$
$$P_4^0(x) = \frac{1}{8}(35x^4-30x^2+3)$$
$$P_4^1(x) = -\frac{5}{2}(7x^3-3x)(1-x^2)^{1/2}$$
$$P_4^2(x) = \frac{15}{2}(7x^2-1)(1-x^2)$$
$$P_4^3(x) = -105x(1-x^2)^{3/2}$$
$$P_4^4(x) = 105(1-x^2)^2$$

★NOTA $(2l)!! = 2l\,(2l-2)(2l-4)(2l-6)\cdots$
$= 2l\,2(l-1)2(l-2)2(l-3)\cdots$
$= 2^l l!$

- **Ortogonalidad**

$\forall m$ fijo,

$$\int_{-1}^{1} P_l^m(x)\, P_{l'}^m(x)\, dx = \frac{2}{2l+1}\frac{(l+m)!}{(l-m)!}\,\delta_{ll'}$$

tal que

$$P_l^{mN}(x) = \sqrt{\frac{2l+1}{2}\frac{(l-m)!}{(l+m)!}}\,P_l^m(x) \quad \text{es una base ortonormal}$$

$\forall l$ fijo,

$$\int_{-1}^{1} \frac{1}{(1-x^2)}\, P_l^m(x)\, P_l^{m'}(x)\, dx = \frac{(l+m)!}{m\,(l-m)!}\,\delta_{mm'}$$

tal que

$$P_l^{mN}(x) = \sqrt{\frac{m\,(l-m)!}{(l+m)!}}\,P_l^m(x) \quad \text{es una base ortonormal}$$

Armónicos Esféricos

Composición de los polinomios asociados de Legendre y las funciones de Fourier

$$T_m(\varphi) = \frac{1}{\sqrt{2\pi}} e^{im\varphi} \qquad P_l^m(\theta) = \qquad \rightarrow \begin{cases} \varphi \in [0, 2\pi] \\ \theta \in [0, \pi] \\ m \in \mathbb{N} \end{cases}$$

Tal que

$$Y_l^m(\theta,\varphi) = (-1)^m \sqrt{\frac{(2l+1)}{2\pi}\frac{(l-m)!}{(l+m)!}}\,P_l^m(\cos\theta)\, e^{im\varphi}$$

$$\rightarrow \int Y_l^{m*}(\Omega)\, Y_{l'}^{m'}(\Omega)\, d\Omega = \delta_{ll'}\delta_{mm'}, \text{ con } d\Omega = \sin\theta\, d\theta\, d\varphi \text{ y } \Omega = \Omega(\theta,\varphi)$$

Los armónicos esféricos forman una base completa

$$f(\theta,\varphi) = \sum_{l=0}^{\infty}\sum_{m=-l}^{l} A_{lm}\, Y_l^m(\theta,\varphi)$$

$$\rightarrow A_{lm} = \int Y_l^{m*}(\theta,\varphi)\, f(\theta,\varphi)\, d\theta\, d\varphi$$

$$\therefore f(\Omega) = \sum_l\sum_m \left(\int Y_l^{m*}(\Omega')\, f(\Omega')\, d\Omega'\right) Y_l^m(\Omega) = \int \left[\sum_{lm} Y_l^{m*}(\Omega') Y_l^m(\Omega)\right] f(\Omega')\, d\Omega'$$

$$= \int f(\Omega')\,\delta(\Omega'-\Omega)\, d\Omega'$$

tal que

$$\sum_{lm} Y_l^{m*}(\Omega')\, Y_l^m(\Omega) = \delta(\Omega'-\Omega) = \delta(\varphi'-\varphi)\,\frac{1}{\sin\theta}\,\delta(\theta'-\theta) = \delta(\varphi'-\varphi)\,\delta(\cos\theta'-\cos\theta)$$

- **Propiedades**

1. $Y_l^{m*}(\Omega) = (-1)^m Y_l^{-m}(\Omega)$ → demostrar usando T_m^* y P_l^{-m}
2. $Y_l^m(\pi-\theta, \varphi+2\pi) = (-1)^l Y_l^m(\theta,\varphi)$ → inversión respecto al origen
3. $Y_l^0(\theta) = \sqrt{\frac{2l+1}{4\pi}}\,P_l(\cos\theta)$
4. $Y_l^m(0,\varphi) = \sqrt{\frac{2l+1}{4\pi}}\,\delta_{m0}$

Teorema de Adición

$$P_l(\cos\gamma) = \frac{4\pi}{2l+1}\sum_{m=-l}^{l} Y_l^{m*}(\theta_1,\varphi_1)\, Y_l^m(\theta_2,\varphi_2)$$

$$Y_0^0(\theta,\varphi) = \frac{1}{2}\sqrt{\frac{1}{\pi}}$$

$$Y_1^{-1}(\theta,\varphi) = \frac{1}{2}\sqrt{\frac{3}{2\pi}}\,\sin\theta\, e^{-i\varphi}$$

$$Y_1^0(\theta,\varphi) = \frac{1}{2}\sqrt{\frac{3}{\pi}}\,\cos\theta$$

$$Y_1^1(\theta,\varphi) = \frac{-1}{2}\sqrt{\frac{3}{2\pi}}\,\sin\theta\, e^{i\varphi}$$

$$Y_2^{-2}(\theta,\varphi) = \frac{1}{4}\sqrt{\frac{15}{2\pi}}\,\sin^2\theta\, e^{-2i\varphi}$$

$$Y_2^{-1}(\theta,\varphi) = \frac{1}{2}\sqrt{\frac{15}{2\pi}}\,\sin\theta\,\cos\theta\, e^{-i\varphi}$$

$$Y_2^0(\theta,\varphi) = \frac{1}{4}\sqrt{\frac{5}{\pi}}\,(3\cos^2\theta - 1)$$

$$Y_2^1(\theta,\varphi) = \frac{-1}{2}\sqrt{\frac{15}{2\pi}}\,\sin\theta\,\cos\theta\, e^{i\varphi}$$

$$Y_2^2(\theta,\varphi) = \frac{1}{4}\sqrt{\frac{15}{2\pi}}\,\sin^2\theta\, e^{2i\varphi}$$

$$Y_3^0(\theta,\varphi) = \frac{1}{4}\sqrt{\frac{7}{\pi}}\,(5\cos^3\theta - 3\cos\theta)$$

Funciones de Hermite

Son las soluciones $H_n(x)$ a la EDO

$$\ddot{H}(x) - 2x\dot{H}(x) + 2nH(x) = 0 \quad \text{ó} \quad \frac{d}{dx}\left(e^{-x^2}\frac{dH(x)}{dx}\right) + 2ne^{-x^2}H(x) = 0 \rightarrow \text{autoadjunta}$$

Aplicando el método de Frobenius

$$H_n(x) = \sum_{s=0}^{[n/2]} (-1)^s \frac{n!}{(n-2s)!\,s!}(2x)^{n-2s}$$

$$\begin{aligned}
H_0(x) &= 1,\\
H_1(x) &= 2x,\\
H_2(x) &= 4x^2 - 2,\\
H_3(x) &= 8x^3 - 12x,\\
H_4(x) &= 16x^4 - 48x^2 + 12,\\
H_5(x) &= 32x^5 - 160x^3 + 120x,\\
H_6(x) &= 64x^6 - 480x^4 + 720x^2 - 120,\\
H_7(x) &= 128x^7 - 1344x^5 + 3360x^3 - 1680x,\\
H_8(x) &= 256x^8 - 3584x^6 + 13440x^4 - 13440x^2 + 1680,\\
H_9(x) &= 512x^9 - 9216x^7 + 48384x^5 - 80640x^3 + 30240x,\\
H_{10}(x) &= 1024x^{10} - 23040x^8 + 161280x^6 - 403200x^4 + 302400x^2 - 30240.
\end{aligned}$$

- **Función Generadora**

$$G(x,t) = e^{-t^2+2tx} = \sum H_n(x)\frac{t^n}{n!}$$

- **Relaciones de Recurrencia**

1. $H_{n+1} + 2nH_{n-1} - 2xH_n = 0$
2. $2nH_{n-1} - H_n' = 0 \rightarrow H_n'(x) = 2nH_{n-1}(x)$

- **Paridad**

★ $H_{2n}(0) = (-1)^n\frac{(2n)!}{n!}$ y $H_{2n-1}(0) = 0$

$$H_n(-x) = (-1)^n H_n(x)$$

- **Fórmula de Rodrigues**

$$H_n(x) = (-1)^n e^{x^2}\frac{d^n}{dx^n}(e^{-x^2})$$

- **Normalización y Ortogonalidad**

Ya que las funciones no son ortogonales en sí mismas, incluimos un factor integrante

$$\int_{-\infty}^{\infty} e^{-x^2} H_n(x) H_m(x) = 0 \quad \text{para } n \neq m$$

Para hallar el factor de normalización,

$$\int e^{-x^2} G(x,s) G(x,t)\,dx = \sum_m \sum_n \frac{s^m t^n}{m!\,n!}\int e^{-x^2} H_m(x) H_n(x)\,dx$$

$$\int e^{-(x^2+s^2-2sx+t^2-2tx)}\,dx = \sum_m \sum_n \frac{s^m t^n}{m!\,n!}\int e^{-x^2} H_m(x) H_n(x)\,dx$$

$$e^{2st}\sqrt{\pi} = \sum_m \sum_n \frac{s^m t^n}{m!\,n!}\int e^{-x^2} H_m(x) H_n(x)\,dx$$

$$\sqrt{\pi}\sum \frac{(2st)^n}{n!} = \sum_m \sum_n \frac{s^m t^n}{m!\,n!}\int e^{-x^2} H_m(x) H_n(x)\,dx \rightarrow \sqrt{\pi}\sum \frac{2^n}{n!}s^n t^n = \sum_m \sum_n \frac{s^m t^n}{m!\,n!}\int e^{-x^2} H_m(x) H_n(x)\,dx$$

Comparando término a término

$$\sqrt{\pi}\sum_n \frac{(2st)^n}{n!} = \sum_n \frac{s^n t^n}{(n!)^2}\int dx\, e^{-x^2}\left[H_n(x)\right]^2 \rightarrow \sqrt{\pi}\,\frac{2^n}{n!} = \frac{1}{(n!)^2}\int dx\, e^{-x^2}\left[H_n(x)\right]^2$$

de donde $\int_{-\infty}^{\infty} dx\, e^{-x^2}\left[H_n(x)\right]^2 = \sqrt{\pi}\,2^n n! \rightarrow H_n^N(x) = \sqrt{\frac{1}{\sqrt{\pi}\,2^n n!}}\,e^{-\frac{x^2}{2}}H_n(x)$

★ De la representación ortonormal,

$$\delta(x) = \sum_0^{\infty} \frac{1}{\sqrt{\pi}\,2^n n!}H_n(0)\,e^{-\frac{x^2}{2}}H_n(x)$$

- **Operadores Escalera**

$$H_{n+1}^N(x) = \frac{1}{\sqrt{2(n+1)}}\left(x - \frac{d}{dx}\right)H_n^N(x) = \hat{N}_+[n]\,H_n^N(x)$$

$$H_{n-1}^N(x) = \frac{1}{\sqrt{2n}}\left(x + \frac{d}{dx}\right)H_n^N(x) = \hat{N}_-[n]\,H_n^N(x)$$

Polinomios de Laguerre

Partimos de

$$x\ddot{y}(x) + (1-x)\dot{y}(x) + n\,y(x) = 0 \quad \rightarrow \text{NO es autoadjunta}$$

$$\rightarrow \frac{d}{dx}\left[xe^{-x}\dot{y}(x)\right] + ne^{-x}y(x) = 0 \;, \quad x \in [0,\infty]$$

Solucionando por Frobenius,

$$\mathcal{L}_n(x) = \sum_{m=0}^{n} \frac{(-1)^m}{m!}\binom{n}{m}x^m \qquad \bigstar \binom{n}{m} = \frac{n!}{m!\,(n-m)!}$$

- **Fórmula de Rodrigues**

$$\mathcal{L}_n(x) = \frac{e^x}{n!}\frac{d^n}{dx^n}(x^n e^{-x})$$

- **Función Generadora**

$$G(x,t) = \frac{e^{-x\frac{t}{1-t}}}{1-t} = \sum \mathcal{L}_n(x)\,t^n$$

- **Relaciones de Recurrencia**

1. $(n+1)\mathcal{L}_{n+1}(x) = (2n+1-x)\mathcal{L}_n(x) - n\mathcal{L}_{n-1}(x)$
2. $x\mathcal{L}_n'(x) = n[\mathcal{L}_n(x) - \mathcal{L}_{n-1}(x)]$

- **Ortogonalidad**

$$\int e^{-x}[\mathcal{L}_n(x)]^2\,dx = 1 \quad \rightarrow \text{Ya está normalizado}$$

n	$L_n(x)$
0	1
1	$-x+1$
2	$(x^2-4x+2)/2$
3	$(-x^3+9x^2-18x+6)/6$
4	$(x^4-16x^3+72x^2-96x+24)/24$
5	$(-x^5+25x^4-200x^3+600x^2-600x+120)/120$
6	$(x^6-36x^5+450x^4-2400x^3+5400x^2-4320x+720)/720$

$\bigstar \mathcal{L}_n(0) = 1$

Polinomios asociados de Laguerre

$$x\ddot{y}$$

$$\rightarrow \mathcal{L}_n^k(x) = \sum (-1)^m \frac{(n+k)!}{(n-m)!(k-m)!}\frac{1}{m!}x^m$$

- **Función Generadora**

$$G_k(x,t) = \frac{e^{-x\frac{t}{1-t}}}{(1-t)^{k+1}} = \sum \mathcal{L}_n^k t^n$$

- **Relaciones de Recurrencia**

1. $(n+1)\mathcal{L}_{n+1}^k = (2n+k+1-x)\mathcal{L}_n^k - (n+k)\mathcal{L}_{n-1}^k$
2. $x\frac{d\mathcal{L}_n^k}{dx} = n\mathcal{L}_n^k - (n+k)\mathcal{L}_{n-1}^k$

- **Ortogonalidad**

Usamos factor de peso $x^k e^{-x}$, tal que

$$L_n^k(x) = \sqrt{\frac{n!}{\Gamma(n+k+1)}}\,\mathcal{L}_n^k(x)$$

define una base ortonormal